

AI-POWERED SIEM ALERT INVESTIGATION CHATBOT USING WAZUH AND SPLUNK

Technical Project Report

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Project Domain:

Cybersecurity | Security Information and Event Management (SIEM)

Technologies:

Python | Flask | Wazuh | Splunk | Ollama | Llama 3 | SQLite

Year:2026

CERTIFICATION

This is to certify that the technical project report entitled

"AI-Powered SIEM Alert Investigation Chatbot using Wazuh and Splunk"

is a record of the work carried out by

Gagana Kuruba

as part of a cybersecurity project implementation.

The project demonstrates the integration of Security Information and Event Management (SIEM)

platforms with Artificial Intelligence techniques for automated security alert investigation

and incident analysis.

Date:22-07-2026

Place: Bengaluru

Signature:K Gagana

Project Author

DECLARATION

I hereby declare that the project report entitled

"AI-Powered SIEM Alert Investigation Chatbot using Wazuh and Splunk"

is my original work and has been developed for the purpose of demonstrating practical implementation of cybersecurity monitoring, SIEM integration, and AI-assisted security analysis.

The information presented in this report is based on the implementation, testing, and analysis performed during the project development.

Date: 22-07-2026

Place: Bengaluru

Signature: K Gagana

Gagana Kuruba

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and guided me throughout the development of this cybersecurity project.

I would like to acknowledge the open-source communities behind Wazuh, Splunk, Flask, Ollama, and other technologies used in this project, whose documentation and resources helped in successful implementation.

I also thank the cybersecurity community for providing valuable knowledge and resources that contributed to improving my understanding of Security Information and Event Management (SIEM), threat detection, and AI-assisted incident investigation.

Finally, I would like to thank my family and friends for their continuous encouragement and support throughout the project development process.

Gagana Kuruba

ABSTRACT

Security Information and Event Management (SIEM) systems are widely used to collect, monitor, and analyze security events generated from various devices and applications within an organization's network. Although SIEM solutions such as Wazuh and Splunk provide centralized monitoring and alert generation, security analysts are often required to manually investigate alerts, correlate security events, assess risks, and determine appropriate response actions. This manual investigation process is time-consuming and may delay incident response during cyberattacks.

The proposed project, **"AI-Powered SIEM Alert Investigation Chatbot using Wazuh and Splunk,"** aims to improve Security Operations Center (SOC) efficiency by integrating artificial intelligence with SIEM technologies. The system collects security alerts from Wazuh and supports integration with Splunk through its REST API, providing centralized visibility of security events. The collected alerts are processed using an alert parser, risk assessment engine, threat intelligence

module, MITRE ATT&CK mapping, and automated incident generation engine.

The application is developed using the Flask framework and provides an interactive SOC dashboard that displays security alerts, active incidents, severity

distribution, MITRE ATT&CK statistics, and overall security analytics. An AI- powered chatbot based on the Llama 3 Large Language Model, executed locally

using Ollama, enables analysts to investigate alerts through natural language queries, generate incident summaries, evaluate risks, and recommend mitigation strategies. In addition, an AI investigation module automatically produces detailed investigation reports for selected security alerts.

The proposed system reduces manual effort in security monitoring, improves the speed of threat investigation, assists SOC analysts in decision-making, and

enhances overall incident response capabilities. The project demonstrates the practical application of artificial intelligence in modern cybersecurity by combining

SIEM technologies, threat intelligence, automated risk assessment, and intelligent security analysis within a unified platform.

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Abbreviations

Abbreviation	Meaning
SIEM	Security Information and Event Management
SOC	Security Operations Center
AI	Artificial Intelligence
FIM	File Integrity Monitoring
API	Application Programming Interfa
LLM	Large Language Model
IDS	Intrusion Detection System
JSON	JavaScript Object Notation
REST	Representational State Transfer
CSV	Comma-Separated Values
PDF	Portable Document Format

CHAPTER 1 INTRODUCTION

1.1 Introduction

The rapid growth of information technology has significantly increased the number of cyber threats targeting organizations across the world. Modern enterprise environments generate enormous volumes of security logs from operating systems, network devices, applications, firewalls, cloud services, and endpoint devices. Monitoring these logs manually is a challenging task due to the increasing complexity and sophistication of cyberattacks. Security analysts working in Security Operations Centers (SOCs) must continuously investigate thousands of security alerts every day, making incident detection and response both time-consuming and resource-intensive.

Security Information and Event Management (SIEM) solutions have become an essential component of modern cybersecurity infrastructure. SIEM platforms

collect, normalize, correlate, and analyze security events from multiple sources to identify suspicious activities and security incidents. Wazuh is an open-source SIEM

platform that provides host intrusion detection, file integrity monitoring, vulnerability assessment, log analysis, and security monitoring. Splunk is another

widely adopted SIEM solution that offers powerful search capabilities, log indexing, visualization, and advanced security analytics.

Although SIEM platforms effectively collect and display security events, security analysts still spend considerable time manually investigating alerts, evaluating risks,

identifying Indicators of Compromise (IOCs), mapping attacks to the MITRE ATT&CK framework, and determining appropriate mitigation strategies. This manual

investigation process often delays incident response and increases analyst workload.

To address these challenges, this project proposes an **AI-Powered SIEM Alert Investigation Chatbot using Wazuh and Splunk**. The system integrates artificial

intelligence with SIEM technologies to automate security alert investigation and assist SOC analysts during incident response. Security alerts collected from Wazuh

and supported through Splunk integration are processed using multiple analysis modules including an Alert Parser, Risk Assessment Engine, Threat Intelligence

Module, MITRE ATT&CK Mapper, Incident Generation Engine, and Automated Response Engine.

The proposed system is developed using the Flask web framework and provides an interactive Security Operations Center dashboard for visualizing alerts, incidents, security analytics, severity distributions, and MITRE ATT&CK statistics. An AI chatbot powered by the Llama 3 Large Language Model running locally through Ollama enables analysts to perform natural language-based investigations, obtain incident summaries, assess risks, and receive recommended response actions. By combining SIEM technologies with artificial intelligence, the system improves threat investigation efficiency, reduces manual effort, and enhances overall cybersecurity operations.



1.2 Existing System

Traditional Security Operations Centers rely on Security Information and Event Management systems to collect and analyze security events generated by various network devices and endpoints. SIEM platforms such as Wazuh and Splunk efficiently aggregate security logs and generate alerts whenever suspicious activities are detected. However, the investigation process is largely manual and depends on the expertise of security analysts.

In conventional systems, analysts are required to review individual alerts, correlate related events, identify affected assets, verify threat intelligence

information, perform risk assessment, and determine appropriate response actions manually. This process consumes significant time, especially in

environments where thousands of alerts are generated daily. Furthermore, repeated manual investigations increase analyst fatigue and may result in delayed responses or overlooked security incidents.

Although existing SIEM solutions provide dashboards, visualization, and search capabilities, they generally do not provide intelligent investigation assistance or

automated incident summarization. As a result, organizations require additional tools and significant human effort to complete security investigations efficiently.

1.3 Proposed System

The proposed system integrates artificial intelligence with SIEM technologies to automate the investigation of security alerts and improve the efficiency of Security Operations Centers. Security events collected by Wazuh are processed through multiple analytical modules including alert parsing, threat intelligence enrichment, risk assessment, MITRE ATT&CK mapping, incident generation, and automated response recommendation. The system is also designed to support integration with Splunk through its REST API for centralized log analysis.

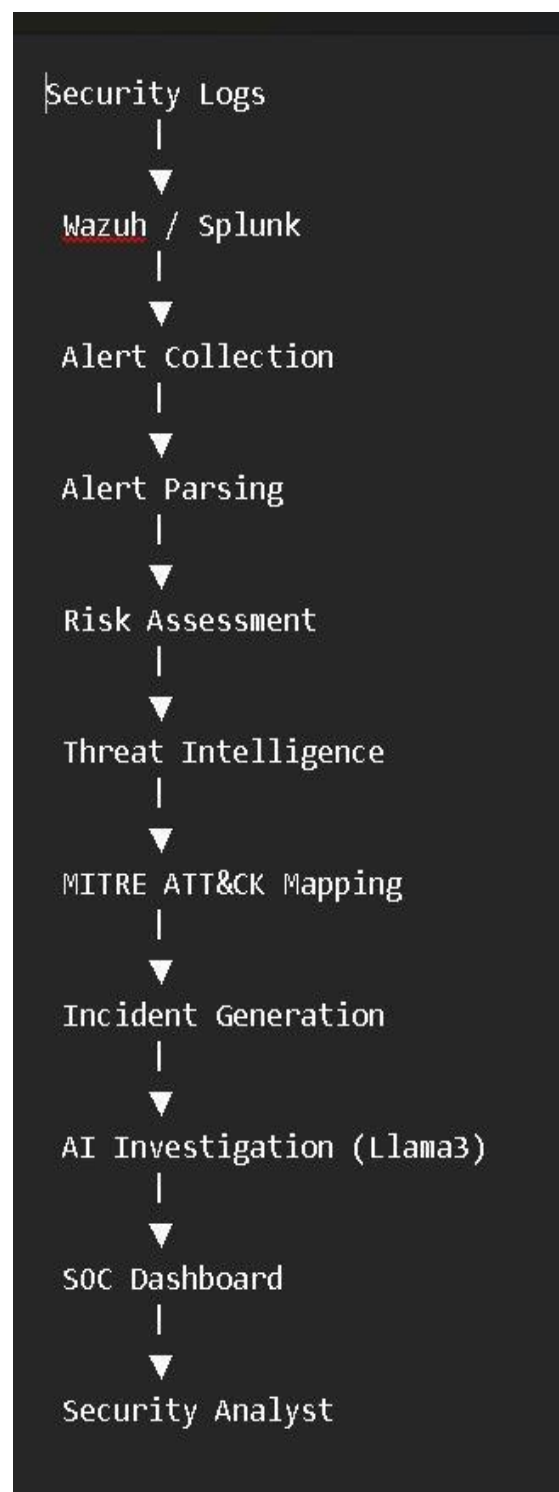
An interactive web dashboard developed using the Flask framework enables analysts to monitor security alerts, active incidents, severity distributions, and

security analytics in real time. An AI-powered chatbot based on the Llama 3 Large Language Model assists analysts by answering cybersecurity queries, generating

incident summaries, evaluating risks, and recommending investigation and mitigation procedures.

The proposed solution reduces manual effort, accelerates security investigations, improves decision-making, and enhances the overall effectiveness of incident response within Security Operations Centers.

Figure 1.2: Workflow of the proposed AI-powered SIEM alert investigation syste



1.4 Objectives

- Develop a centralized SIEM dashboard for security monitoring.
- Collect and analyze alerts generated by Wazuh.
- Support integration with Splunk using REST APIs.
- Automate risk assessment and incident generation.
- Map alerts to the MITRE ATT&CK framework.
- Integrate threat intelligence for IP reputation analysis.
- Develop an AI chatbot for SOC analysts using Llama 3 and Ollama.
- Reduce manual effort and improve incident response time.

1.5 Advantages

- Centralized security monitoring. Faster
- alert investigation.
- AI-assisted incident analysis.
- Automated risk scoring.
- MITRE ATT&CK mapping.
- Threat intelligence integration.
- Interactive SOC dashboard.
- Reduced analyst workload.
- Improved incident response efficiency.

1.6 Organization of the Report

Briefly describe each chapter, for example:

- **Chapter 1** introduces the project and its objectives.
- **Chapter 2** presents the project description and scope.
- **Chapter 3** discusses system analysis and requirements. •
- Chapter 4** explains the system design and architecture. • **Chapter 5** describes the implementation details.
- **Chapter 6** presents the results, screenshots, and evaluation.
- The final chapters summarize the conclusion, future scope, and references.

CHAPTER 2 PROJECT

DESCRIPTION

2.1 Problem Definition

Modern organizations generate thousands of security events every minute from servers, workstations, firewalls, applications, cloud services, and network devices. Security Information and Event Management (SIEM) platforms such as Wazuh and Splunk collect and centralize these security logs, enabling organizations to monitor potential cyber threats. However, despite the capabilities provided by SIEM platforms, security analysts are still required to manually investigate alerts before determining whether they represent genuine security incidents.

Manual investigation involves reviewing event logs, identifying the affected assets, performing risk assessment, mapping attacks to the MITRE ATT&CK framework, verifying threat intelligence, and recommending appropriate mitigation measures. As the number of alerts increases, Security Operations Center (SOC) analysts experience alert fatigue, leading to delayed incident response and the possibility of overlooking critical security threats.

Although SIEM solutions provide centralized log collection and visualization, they do not automatically perform intelligent investigation or generate comprehensive incident reports. Organizations therefore require an automated solution capable of assisting analysts in understanding alerts, prioritizing incidents, performing threat analysis, and recommending response actions.

The proposed AI-Powered SIEM Alert Investigation Chatbot addresses these challenges by integrating artificial intelligence with SIEM technologies. The system automates alert investigation using Wazuh, supports Splunk integration through REST APIs, enriches alerts using threat intelligence, performs risk assessment, maps attacks to the MITRE ATT&CK framework, and generates AI-assisted investigation reports through a conversational chatbot interface.

2.2 Project Description

The AI-Powered SIEM Alert Investigation Chatbot using Wazuh and Splunk is designed to improve Security Operations Center efficiency by automating various stages of security alert investigation. The application integrates security monitoring, artificial intelligence, threat intelligence, and incident management into a unified platform.

The system retrieves security alerts from the Wazuh Manager through its REST API and is designed to support integration with Splunk Enterprise using the Splunk

REST API. Each alert is processed through an Alert Parser that extracts relevant information such as event description, source IP address, hostname, severity level,

timestamp, and MITRE ATT&CK mapping.

After parsing, the Risk Assessment Engine evaluates the severity of each alert based on predefined rules and assigns an appropriate risk score. The Threat

Intelligence module enriches alerts by checking IP reputation and identifying potentially malicious sources. The Incident Engine automatically groups related

alerts into incidents, allowing analysts to investigate correlated security events instead of individual alerts.

The Flask-based web dashboard provides a centralized interface for monitoring alerts, viewing active incidents, analyzing severity distributions, examining MITRE

ATT&CK mappings, and reviewing security analytics. The dashboard also displays graphical representations of security statistics to assist analysts in identifying

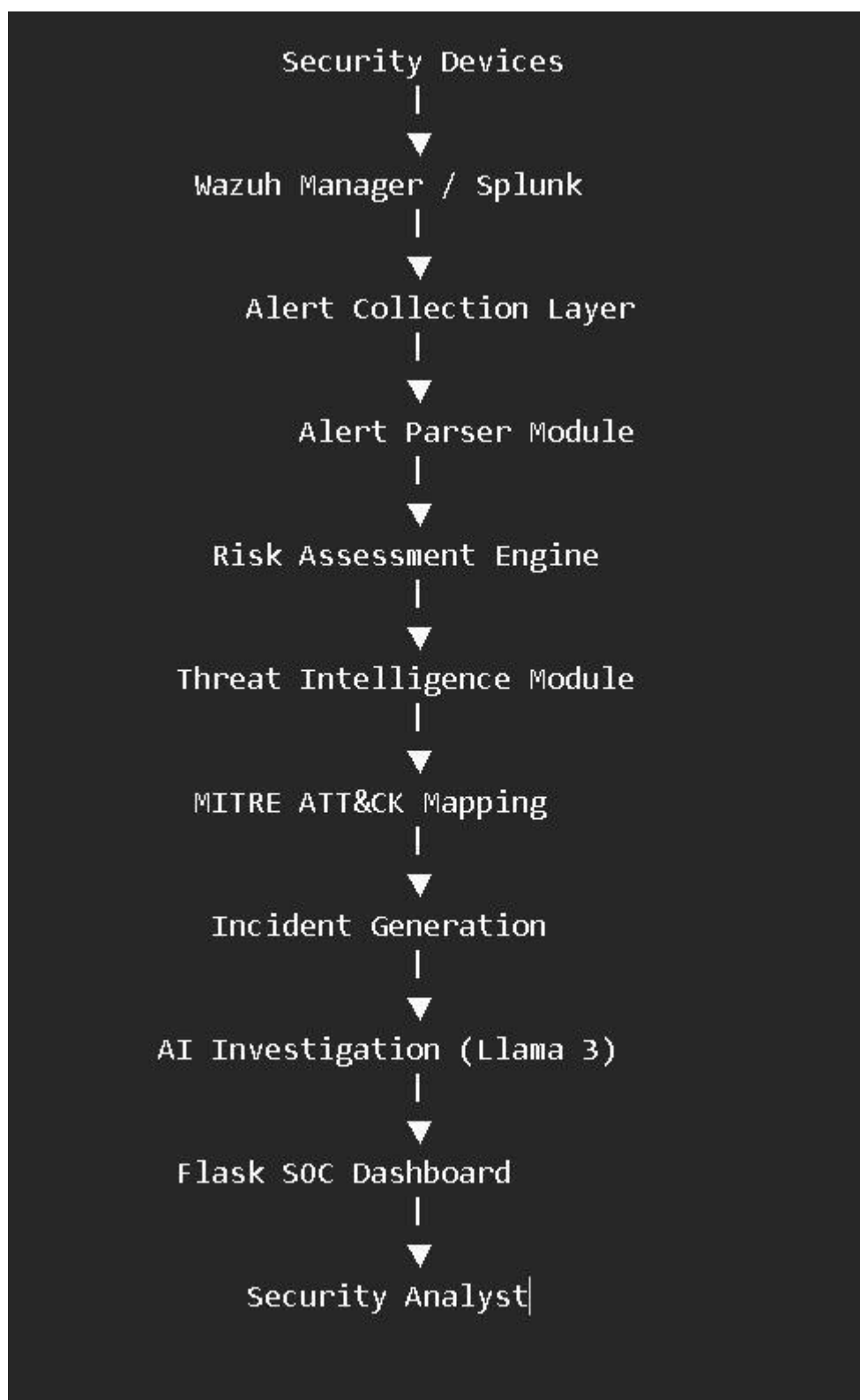
attack trends.

An AI chatbot powered by the Llama 3 Large Language Model running locally through Ollama enables analysts to interact with the system using natural

language. The chatbot provides incident summaries, risk assessments, investigation guidance, and recommended mitigation strategies, thereby reducing

manual effort and improving SOC productivity.

Figure 2.1 – Project Workflow



2.3 Scope of the Project

The scope of this project is to design and develop an intelligent Security Operations Center platform that assists cybersecurity analysts in monitoring and investigating security alerts efficiently. The system focuses on integrating Security Information and Event Management technologies with artificial intelligence to improve threat detection and incident response.

The project includes the collection of security alerts from Wazuh and supports integration with Splunk through REST APIs. It provides automated alert parsing,

risk assessment, threat intelligence enrichment, MITRE ATT&CK mapping, incident generation, and response recommendation. A web-based dashboard enables

analysts to visualize alerts, monitor active incidents, and analyze security trends through charts and statistics.

The AI-powered chatbot provides conversational assistance by answering cybersecurity-related questions, generating incident summaries, evaluating risks,

and recommending mitigation procedures. The system is intended for educational and research purposes and demonstrates how artificial intelligence can enhance

Security Operations Center operations.

The project does not replace commercial SIEM solutions or security analysts; instead, it serves as an intelligent assistant that improves investigation efficiency, reduces manual effort, and supports faster decision-making during cybersecurity incident response.

CHAPTER 3 SYSTEM

ANALYSIS

3.1 System Analysis

System analysis is an essential phase of software development that focuses on understanding user requirements, identifying system functionalities, and defining the overall behavior of the proposed solution. It provides a detailed understanding of the existing system, identifies its limitations, and determines the requirements necessary for developing an improved solution.

The proposed AI-Powered SIEM Alert Investigation Chatbot is designed to simplify the investigation of security alerts generated by Security Information and Event

Management (SIEM) platforms. The system analyzes alerts collected from Wazuh and supports integration with Splunk, processes security events using risk assessment and

threat intelligence modules, maps attacks to the MITRE ATT&CK framework, and assists Security Operations Center (SOC) analysts through an AI-powered chatbot.

The analysis phase identified the need for centralized monitoring, automated incident generation, AI-assisted investigation, and improved visualization of security

events. These requirements formed the basis for the development of the proposed intelligent SOC platform.

3.2 Feasibility Study

3.2.1 Technical Feasibility

The proposed system is technically feasible because it is developed using open- source technologies including Python, Flask, Wazuh, Ollama, and SQLite. Splunk integration is supported through REST APIs. The required software components are readily available and compatible with standard Windows or Linux environments. Artificial intelligence functionality is implemented locally using the Llama 3 model executed through Ollama, eliminating dependence on external cloud services.

3.2.2 Economic Feasibility

The project is economically feasible because it utilizes open-source software and freely available development tools. Wazuh, Flask, Python, SQLite, and Ollama significantly reduce software licensing costs. The project can be implemented using existing computer hardware without requiring specialized infrastructure, making it suitable for educational institutions and small organizations.

3.2.3 Operational Feasibility

The proposed system provides a user-friendly web interface that enables Security Operations Center analysts to monitor alerts, investigate incidents, and interact with the AI chatbot efficiently. The dashboard presents security information through tables, charts, and graphical visualizations, reducing the complexity of manual log analysis. The automated investigation process minimizes analyst workload and improves operational efficiency.

3.3 Functional Requirements

The proposed system performs the following functional requirements:

- Collect security alerts from the Wazuh REST API.
- Support security log retrieval from Splunk REST API. •

Parse security alerts into structured information.

- Perform automated risk assessment.
- Integrate threat intelligence for IP reputation analysis.
- Generate active incidents automatically.
- Map security events to the MITRE ATT&CK framework.
- Display security analytics through the SOC dashboard. Provide
- AI-powered investigation reports.
- Support natural language interaction through the AI chatbot.

3.4 Non-Functional Requirements

The system satisfies the following non-functional requirements:

- High availability during security monitoring.
- User-friendly graphical interface.
- Fast response time for dashboard loading.
- Secure communication between system components.
- Reliable storage and processing of security alerts.
- Scalable architecture supporting future enhancements.
- Maintainability through modular software design.
- Compatibility with Windows and Linux operating systems. •

Extensibility for integrating additional SIEM platforms.

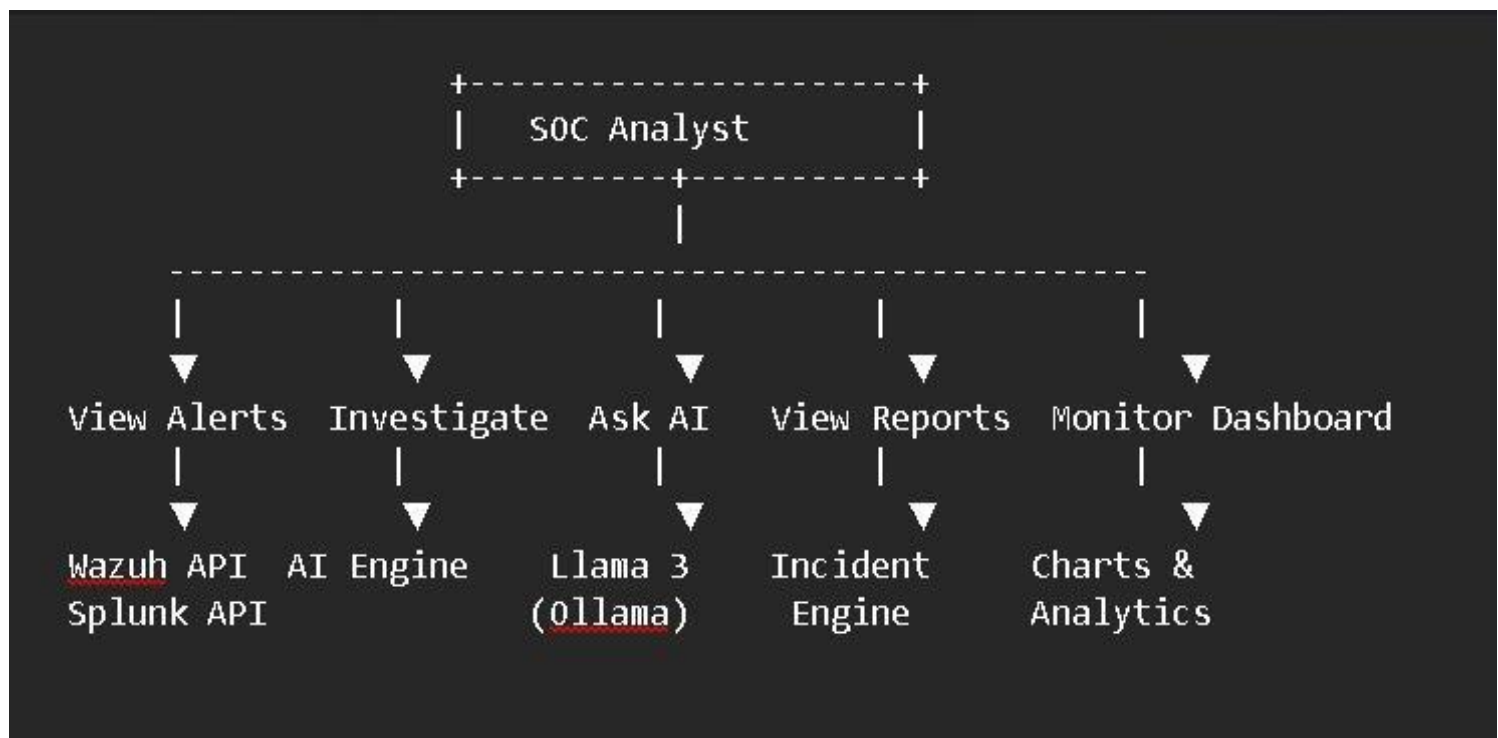
3.5 Hardware Requirements
Specification

Component	
Processor	Intel Core i5 or above
RAM Storage	Minimum 8 GB
Network	100 GB HDD/SSD Internet/LAN
Display	Connection
	1366 × 768 Resolution or above

3.6 Software Requirements

Software	Purpose
Windows 11 / Ubuntu	Operating System
Python 3.11	Programming Language
Flask	Web Framework
Wazuh	SIEM Platform
Splunk Enterprise	Log Analytics Platform
SQLite	Database
Ollama	LLM Runtime
Llama 3	AI Model
Visual Studio Code	Development IDE
Git	Version Control
Chrome / Edge	Web Browser

3.7 Use Case Diagram



3.8 Use Case Description

Actor	Description
SOC Analyst	Monitors and investigates security alerts.
View Alerts	Displays alerts collected from Wazuh and Splunk.
Investigate Alert	Generates AI-assisted investigation reports.
Ask AI	Allows analysts to ask cybersecurity-related questions.
View Reports	Displays incident reports and recommendations.
Dashboard	Visualizes analytics and security statistics.

CHAPTER 4 SYSTEM DESIGN

4.1 System Design

System design defines the overall architecture and internal organization of the proposed AI-Powered SIEM Alert Investigation Chatbot. The objective of the design is to develop a modular, scalable, and efficient Security Operations Center (SOC) platform capable of collecting, analyzing, investigating, and visualizing security alerts generated by SIEM platforms.

The proposed system consists of multiple interconnected modules including Wazuh integration, Splunk integration, Alert Parser, Risk Assessment Engine,

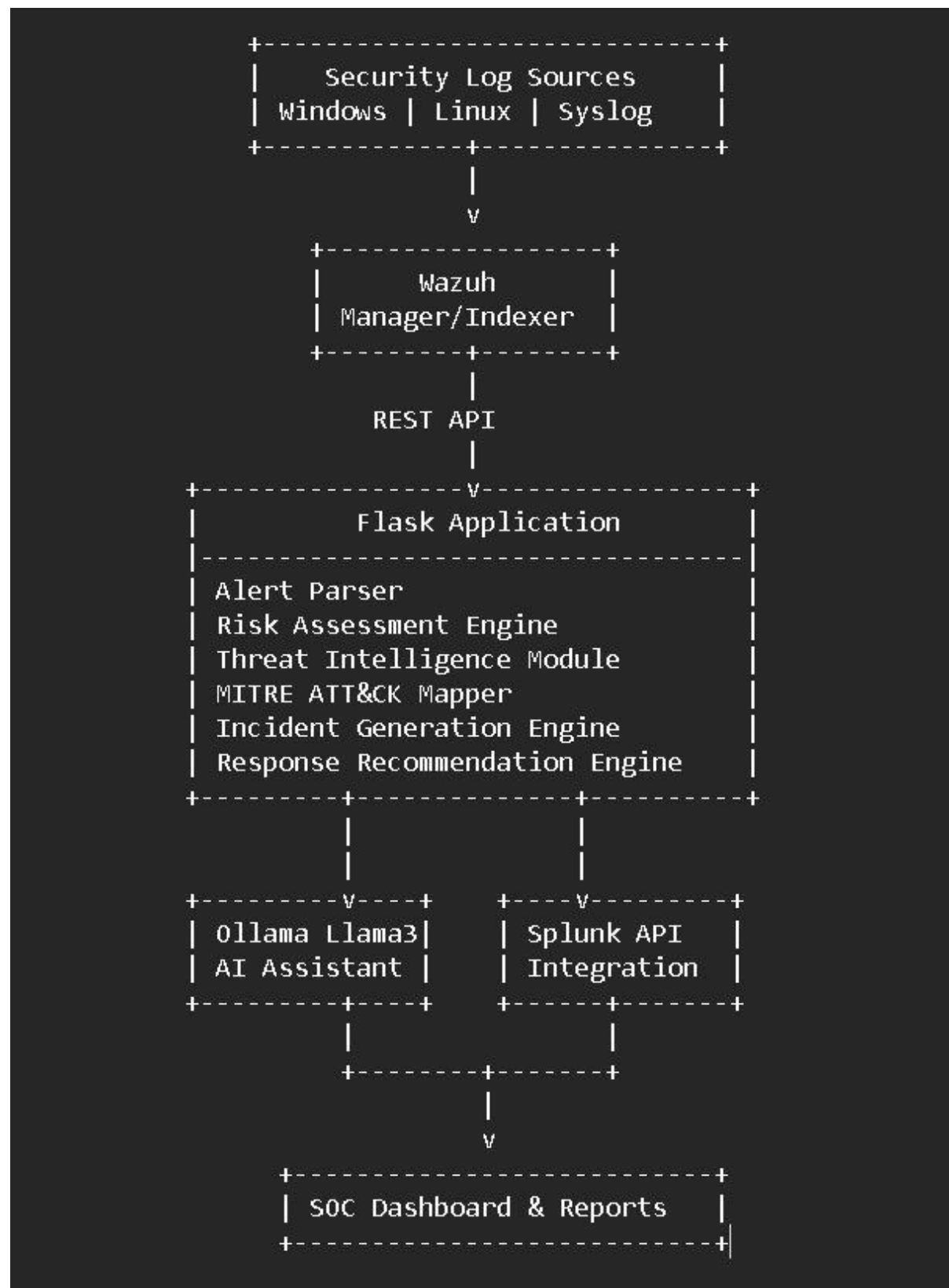
Threat Intelligence Module, MITRE ATT&CK Mapping, Incident Generation Engine, AI Investigation Module, and a Flask-based SOC Dashboard. Each module performs

a specific responsibility while collectively supporting automated security monitoring and incident investigation.

The modular architecture improves maintainability, scalability, and future enhancement by allowing additional SIEM platforms or security tools to be

integrated with minimal modification to the existing system.

4.2 Overall System Architecture



4.3 Module Architecture

The proposed system is divided into several functional modules to simplify development and maintenance. The Alert Collection Module retrieves security events from Wazuh and Splunk. The Alert Parser extracts relevant fields including severity, event description, agent information, IP address, timestamp, and MITRE ATT&CK details.

The Risk Assessment Engine evaluates each alert and assigns a corresponding risk score and risk level. The Threat Intelligence Module enriches alerts by obtaining IP

reputation and geographical information. The MITRE ATT&CK Mapping Module identifies the associated attack techniques and tactics.

The Incident Engine groups related alerts into incidents, enabling analysts to investigate correlated security events rather than isolated alerts. Finally, the AI

Investigation Module interacts with the Llama 3 Large Language Model through Ollama to generate intelligent incident summaries, risk assessments, investigation

guidance, and recommended response actions. All processed information is presented through the Flask-based Security Operations Center dashboard.

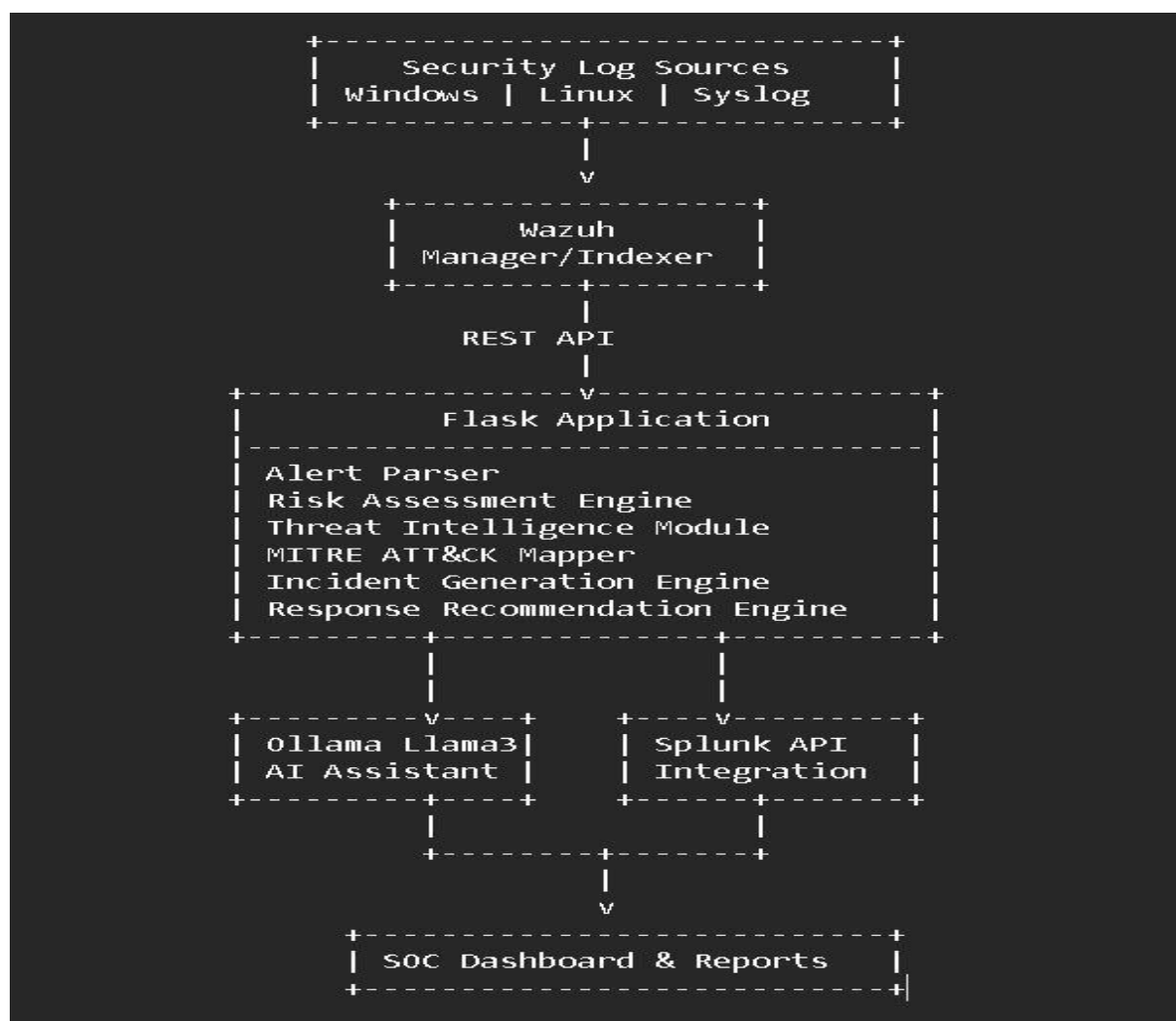


Figure 4.2 Module Architecture



4.4 Data Flow Diagram (DFD)



4.5 Wazuh Architecture

Wazuh serves as the primary SIEM platform in the proposed system. It continuously monitors endpoint devices, collects operating system logs, detects security events, performs vulnerability assessment, and generates security alerts. The Wazuh Manager processes incoming events from agents and stores them within the Wazuh Indexer.

The Flask application retrieves security alerts from the Wazuh REST API, allowing them to be processed further by the alert parser and risk assessment modules.

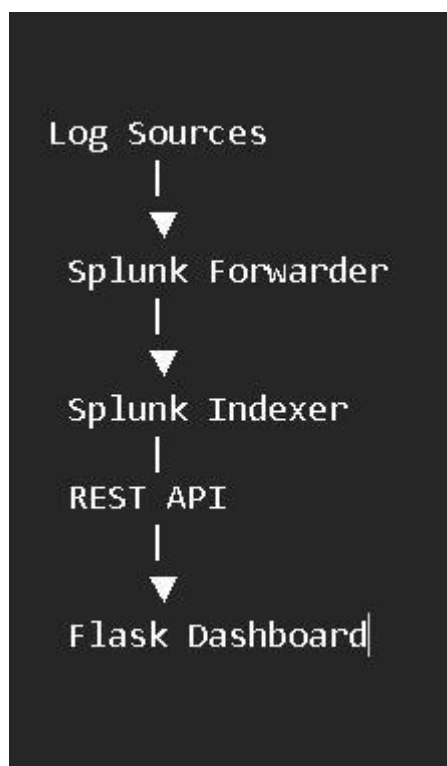
Figure 4.4 Wazuh Architecture



4.6 Splunk Architecture

The proposed system supports integration with Splunk Enterprise using the Splunk REST API. Splunk collects machine-generated data from various sources, indexes the logs, and enables efficient searching and visualization. The Flask application communicates with Splunk through authenticated REST API requests to retrieve security events and display them within the unified SOC dashboard whenever the Splunk service is available.

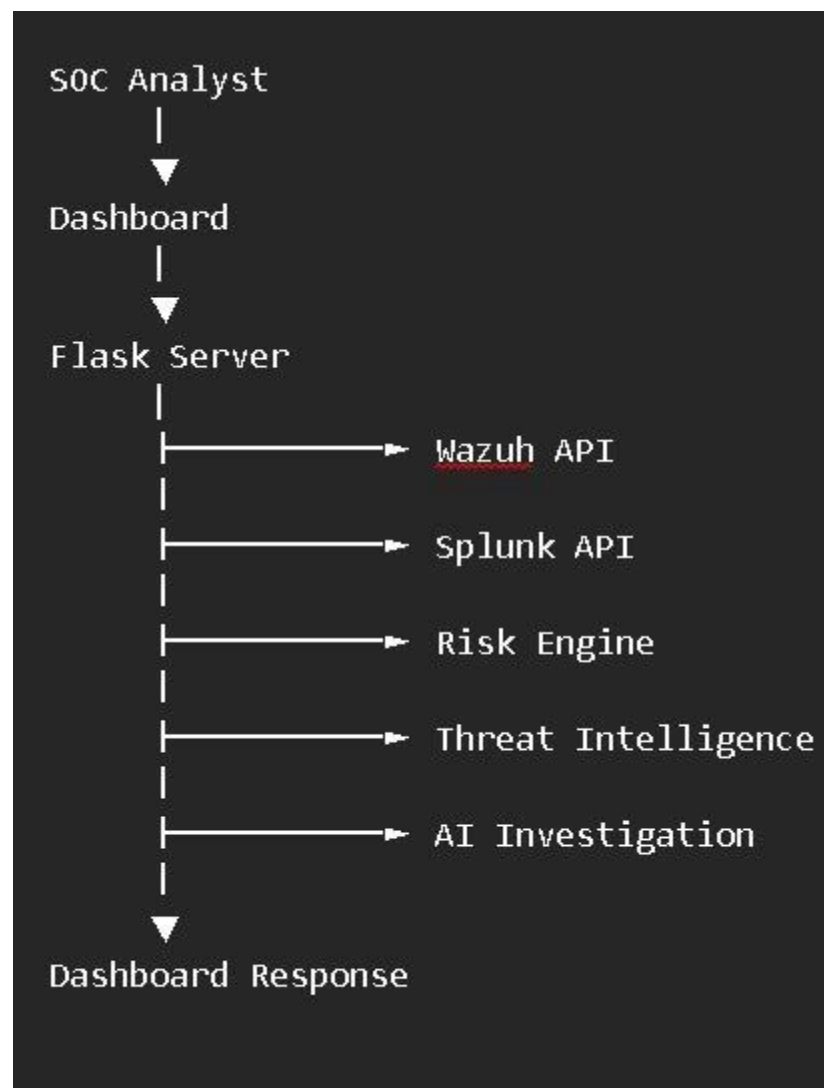
Figure 4.5 Splunk Architecture



4.7 Database Design

Field	Description
Event	Alert description
Severity	Alert severity
Agent	Wazuh agent name
IP Address	Source IP
MITRE	MITRE ATT&CK technique
Risk Score	Calculated risk
Country	Threat intelligence location
Reputation	IP reputation

4.8 Sequence Diagram



CHAPTER 5 IMPLEMENTATION

5.1 Development Environment

The AI-Powered SIEM Alert Investigation Chatbot was developed using Python and the Flask web framework. The application was implemented in a Windows 11 environment using Visual Studio Code as the Integrated Development Environment (IDE). Version control was maintained using Git, while Python virtual environments were used for dependency management.

The system integrates multiple cybersecurity technologies including Wazuh for Security Information and Event Management (SIEM), Splunk for centralized log

analysis, Ollama for executing the Llama 3 Large Language Model locally, and SQLite for lightweight data storage where required. REST APIs are used for

communication between the Flask application and external security platforms.

The implementation follows a modular architecture where each functional component performs an independent task, making the application easier to maintain, extend, and debug.

5.2 Technology Stack

Technology	Purpose
Python	Programming
3.11 Flask	Language Web
HTML5	Framework
CSS3	User Interface
JavaScript	Dashboard Styling
	Client-side
	Interaction
Wazuh REST API	
Splunk REST API	SIEM Alert Collection
Ollama	Log Collection
Llama	Local LLM Runtime
3	AI Security Investigation
SQLite	Database
Requests Library	REST API
Chart.js	Communication
Visual Studio	Dashboard Analytics
Code Git	Development IDE
	Version Control

5.3 Project Modules

5.3.1 Flask Web Application

The Flask application serves as the central controller of the proposed system. It handles incoming HTTP requests, communicates with backend modules, retrieves security alerts, processes investigation requests, and renders the Security Operations Center dashboard.

The application consists of multiple routes responsible for loading the dashboard, displaying investigation reports, interacting with the AI chatbot, and retrieving

security information from Wazuh and Splunk. Flask provides lightweight routing and template rendering, making it suitable for developing the web-based SOC

dashboard.

5.3.2 Wazuh Integration

Wazuh serves as the primary Security Information and Event Management platform within the proposed system. Security alerts are retrieved from the Wazuh Manager through REST APIs using authenticated HTTP requests. The collected alerts include event descriptions, severity levels, timestamps, agent information, source IP addresses, and MITRE ATT&CK mappings.

The retrieved alerts are forwarded to the Alert Parser module, where they are converted into a structured format for further processing. Wazuh provides

continuous monitoring of endpoint devices and acts as the primary source of security events within the proposed system.

5.3.3 Splunk Integration

The proposed system supports integration with Splunk Enterprise using the Splunk REST API. The Flask application sends authenticated REST requests to retrieve indexed security events from Splunk whenever the service is available. Splunk integration enables centralized log analysis and allows the application to correlate security events collected from multiple sources.

If Splunk is unavailable or unreachable, the application continues operating using Wazuh alerts, thereby ensuring uninterrupted dashboard functionality and maintaining system reliability.

5.3.4 Alert Parser

The Alert Parser extracts meaningful information from raw Wazuh and Splunk alerts. It retrieves attributes such as event description, severity, agent name, timestamp, IP address, hostname, MITRE ATT&CK information, and other security metadata.

The parser converts heterogeneous security logs into a standardized internal representation that is used throughout the application. This standardization

simplifies subsequent processing by the Risk Assessment Engine, Threat Intelligence Module, and AI Investigation module.

5.3.5 Risk Assessment Engine

The Risk Assessment Engine evaluates the severity of each security alert and assigns a corresponding numerical risk score and qualitative risk level. The assessment considers multiple alert attributes including event severity, attack characteristics, and security context.

The calculated risk values enable analysts to prioritize high-risk incidents, improving incident response efficiency and reducing investigation time.

5.3.7 Incident Generation Engine

The Incident Generation Engine automatically groups related security alerts into logical incidents. Similar alerts sharing common characteristics such as event type, affected host, or severity are consolidated into a single incident. This approach reduces duplicate investigations and allows analysts to focus on security incidents instead of individual alerts.

5.3.8 Response Recommendation Engine

The Response Recommendation Engine provides appropriate mitigation strategies based on the characteristics of each security alert. Recommended actions may include monitoring suspicious activities, investigating affected systems, notifying the Security Operations Center, or isolating compromised hosts depending on the calculated risk level.

5.3.9 AI Chatbot Module

The AI Chatbot is implemented using the Llama 3 Large Language Model executed locally through Ollama. The chatbot enables SOC analysts to interact with the system using natural language queries. It answers cybersecurity-related questions, explains security alerts, summarizes incidents, recommends investigation procedures, and provides mitigation guidance.

The chatbot operates entirely within the local environment, ensuring privacy and eliminating dependence on external cloud-based AI services.

5.3.10 AI Investigation Module

The AI Investigation Module automatically generates detailed investigation reports for selected security alerts. The module receives processed alert information including severity, MITRE ATT&CK mapping, threat intelligence results, and risk assessment before constructing an investigation prompt for the Llama 3 model.

The generated report includes an incident summary, risk analysis, investigation findings, and recommended mitigation strategies, significantly reducing manual analysis effort.

5.4 Dashboard Implementation

The Security Operations Center (SOC) Dashboard serves as the primary interface through which security analysts monitor and investigate cyber security events. The dashboard has been developed using the Flask web framework along with HTML5, CSS3, JavaScript, and Chart.js to provide an interactive and responsive user experience. The design emphasizes simplicity, usability, and real-time visibility of security events.

When the application starts, it automatically retrieves security alerts from the Wazuh REST API and supports the integration of Splunk alerts through the Splunk

REST API. Each alert is processed by the Alert Parser, Risk Assessment Engine, and Threat Intelligence Module before being displayed on the dashboard.

The dashboard presents several information panels including Total Alerts, Wazuh Alerts, Splunk Alerts, and High Severity Alerts. These summary cards allow SOC

analysts to quickly assess the overall security status of the monitored environment.

A dedicated Active Incident section groups related security events into incidents generated by the Incident Engine. Each incident contains an Incident ID, title,

severity level, calculated risk score, investigation status, number of associated alerts, and the assigned SOC analyst. This organization reduces duplicate

investigations and improves incident management.

The Latest Security Alerts table displays detailed information for every collected alert, including the event description, severity level, monitored agent, IP address, MITRE ATT&CK mapping, country information, IP reputation, abuse score, recommended response actions, and investigation options. This table enables analysts to review detailed security information without manually searching through raw logs.

Graphical analytics are implemented using Chart.js. The dashboard includes severity distribution charts, alert source comparisons, and MITRE ATT&CK

visualizations that provide an overview of attack trends within the monitored infrastructure. These visualizations improve situational awareness and support

rapid security decision-making.

Overall, the SOC Dashboard acts as a centralized monitoring platform that combines security monitoring, incident management, AI investigation, and security analytics into a single web application.

5.5 AI Chatbot Implementation

The AI Chatbot is one of the primary features of the proposed system. It enables Security Operations Center analysts to interact with the application using natural language instead of manually searching through security logs. The chatbot is implemented using the Llama 3 Large Language Model executed locally through the Ollama framework.

Whenever a user submits a question through the dashboard, the application first analyzes the query using the Intent Detection module. Frequently requested

information such as the number of Wazuh alerts, available agents, high-severity incidents, MITRE ATT&CK mappings, or Splunk alerts is answered directly using

the application's internal modules.

If the question requires detailed analysis, the application automatically constructs a structured cybersecurity prompt containing the user's question and predefined investigation instructions. This prompt is then transmitted to the locally running Llama 3 model through the Ollama REST API.

The Large Language Model analyzes the prompt and generates a professional cybersecurity response containing an incident summary, risk assessment,

investigation procedure, and recommended mitigation steps. The generated response is displayed immediately on the SOC dashboard.

Since the AI model executes locally, no security information is transmitted to external cloud services. This improves data privacy, reduces operational costs, and

allows the chatbot to function without an internet connection.

The AI chatbot significantly reduces the manual workload of SOC analysts by providing intelligent explanations of security events, answering cybersecurity-related questions, and assisting during incident investigation.

5.6 Security Analytics

Security analytics transforms raw security logs into meaningful information that enables Security Operations Center analysts to identify threats efficiently. After collecting alerts from Wazuh and Splunk, the system processes every alert using multiple analytical modules including Risk Assessment, Threat Intelligence, MITRE ATT&CK Mapping, and Incident Generation.

The dashboard displays the total number of alerts currently available within the system. Separate counters indicate the number of alerts received from Wazuh and

Splunk, allowing analysts to understand the contribution of each SIEM platform. High severity alerts are highlighted separately to ensure they receive immediate attention.

The Risk Assessment Engine classifies alerts into Low, Medium, and High risk categories based on predefined security rules. The severity distribution chart visually presents these classifications, allowing analysts to identify increases in malicious activities.

The Threat Intelligence Module enriches security alerts by retrieving IP reputation, abuse score, geographical location, and malicious activity indicators whenever valid IP addresses are available. This additional context helps analysts distinguish between legitimate and suspicious network activity.

The Incident Generation Engine groups related alerts into incidents according to common characteristics such as event type, host, or severity level. Instead of investigating each alert independently, analysts can investigate correlated incidents, significantly improving operational efficiency.

MITRE ATT&CK mappings extracted from Wazuh alerts are visualized through dedicated charts. These visualizations identify frequently observed attacker tactics and techniques, providing valuable insight into current attack patterns.

Overall, the Security Analytics module enables real-time monitoring, improves threat visibility, supports faster incident prioritization, and provides actionable information through intuitive graphical representations.

5.7 Summary

The implementation phase successfully integrates Security Information and Event Management technologies with artificial intelligence to develop a comprehensive Security Operations Center platform. The proposed system combines Wazuh, Splunk, Flask, Ollama, Llama 3, Threat Intelligence, Risk Assessment, Incident Management, and interactive security analytics within a unified architecture.

Each functional module performs an independent responsibility while contributing to the complete investigation workflow. The Alert Parser standardizes

security events, the Risk Assessment Engine prioritizes incidents, the Threat Intelligence Module enriches alerts, and the Incident Engine correlates related

security events. The AI Investigation Module and AI Chatbot automate security analysis by generating detailed investigation reports and providing natural language interaction for SOC analysts.

The Flask-based dashboard presents security information through interactive tables, analytical charts, incident summaries, and AI-generated recommendations, thereby reducing manual investigation effort and improving the efficiency of cybersecurity operations.

The modular implementation approach adopted in this project ensures high maintainability, scalability, and extensibility. Additional SIEM platforms, threat

intelligence services, and machine learning techniques can be integrated into the existing architecture with minimal modifications. The completed implementation

demonstrates the effectiveness of combining artificial intelligence with modern SIEM technologies to support intelligent cybersecurity monitoring and incident response.

CHAPTER 6

RESULTS AND DISCUSSION

The AI-Powered SIEM Alert Investigation Chatbot was successfully implemented and executed using Python and the Flask web framework. The application communicates with the Wazuh Manager through REST APIs to retrieve security alerts and supports integration with Splunk Enterprise for centralized log analysis. Artificial intelligence functionality is implemented locally using the Llama 3 Large Language Model executed through Ollama.

During execution, the application successfully collected security alerts from monitored endpoints, processed them using the Alert Parser, calculated risk

scores, enriched alerts with threat intelligence, generated active incidents, and displayed all information through the Security Operations Center dashboard.

The implemented system enables SOC analysts to investigate security events efficiently using graphical visualizations, automated incident reports, and AI-

assisted cybersecurity analysis.

6.1 Dashboard Implementation

The Security Operations Center (SOC) dashboard serves as the primary interface of the proposed system. It consolidates security information received from both Wazuh and Splunk into a single centralized view. The dashboard presents critical security metrics including the total number of alerts, high-severity incidents, security score, and the most active attacker IP address. This centralized visualization allows security analysts to quickly assess the current security posture without manually reviewing individual logs.

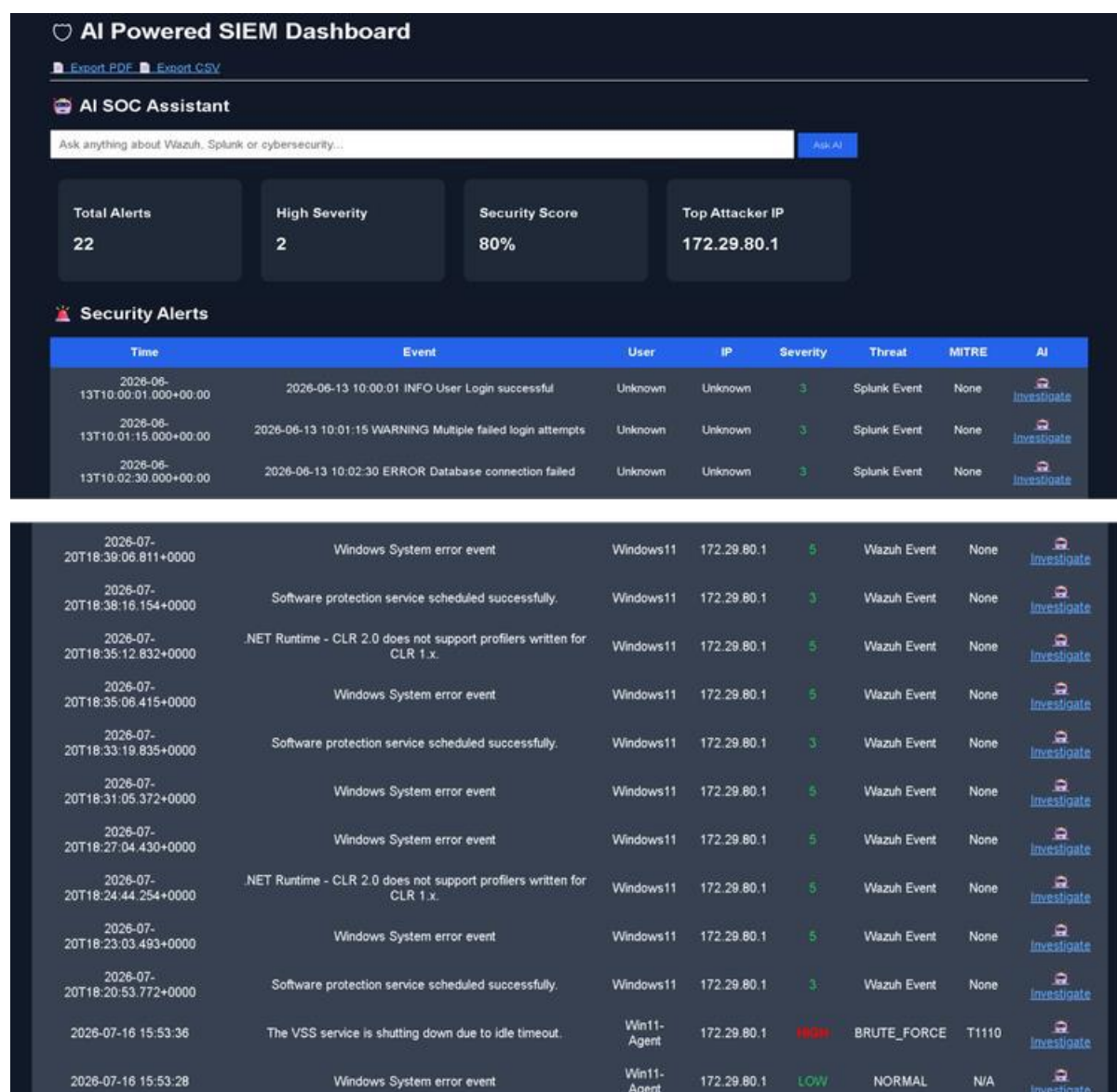


Figure 6.1: AI-Powered SIEM Dashboard displaying real-time security monitoring.

6.2 Wazuh Security Monitoring

The Wazuh platform successfully monitored endpoint activities and generated security events based on Windows system logs and security policies. The dashboard displayed active endpoints, threat hunting information, rule-based detections, server statistics, and security events generated by the monitored system.

The Wazuh Manager continuously collected endpoint logs and classified security events according to predefined detection rules. Built-in MITRE ATT&CK mappings provided additional threat intelligence for security investigations.

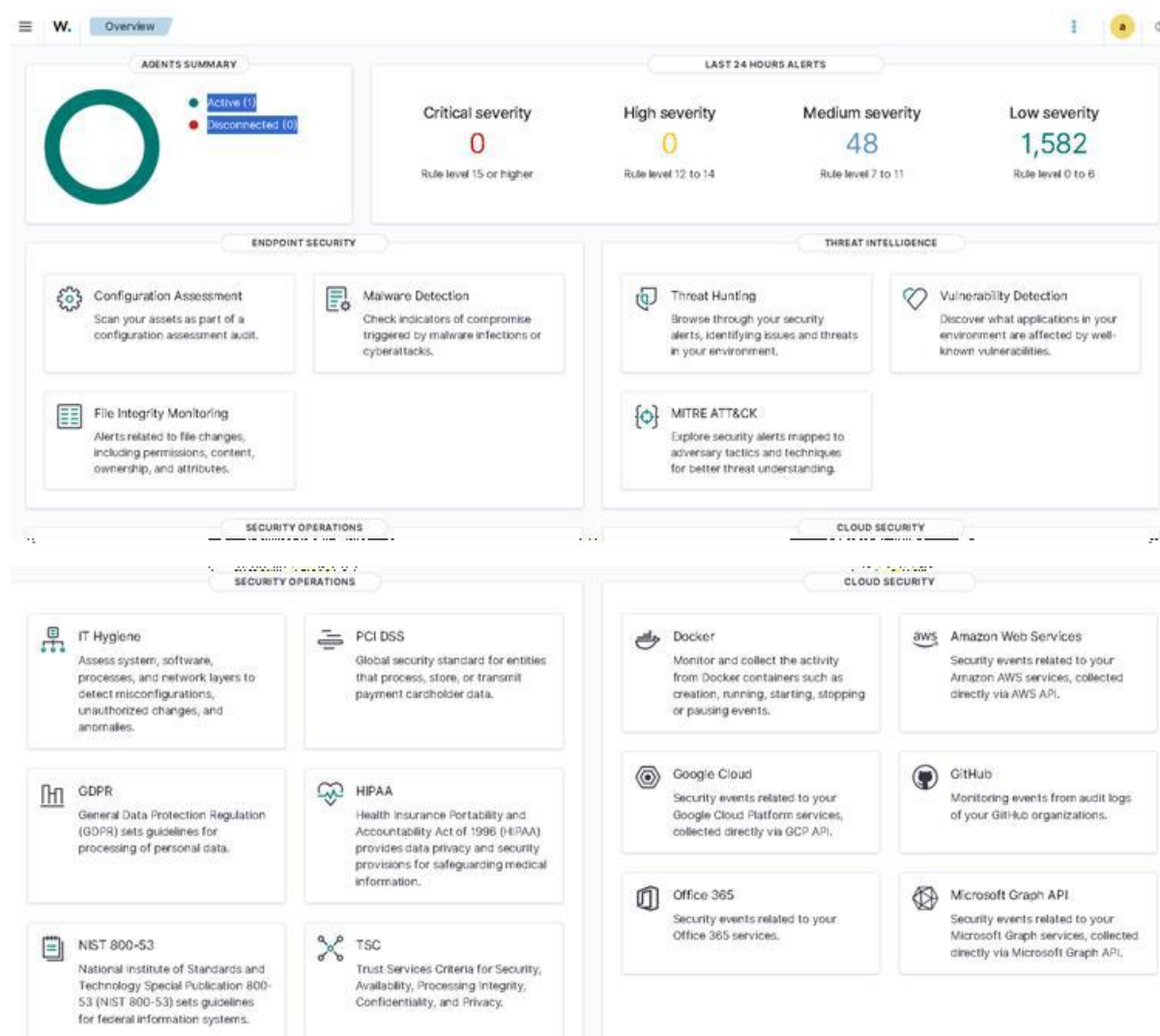


Figure 6.2: Wazuh Dashboard displaying endpoint monitoring and security events.

6.3 AI-Based Security Investigation

The integrated AI chatbot significantly simplified security investigations by allowing analysts to ask cybersecurity questions using natural language. The chatbot successfully interpreted user queries such as:

- Show Agents
- Show Splunk Alerts
- Investigate Alerts
- Explain Security Events

The chatbot combined information collected from both Wazuh and Splunk before generating a cybersecurity-focused response using the locally hosted Llama 3 Large Language Model.

Unlike conventional dashboards that require manual log analysis, the chatbot automatically summarized alerts, identified suspicious behavior, estimated threat severity, and recommended investigation steps.

- [Show Agents](#)

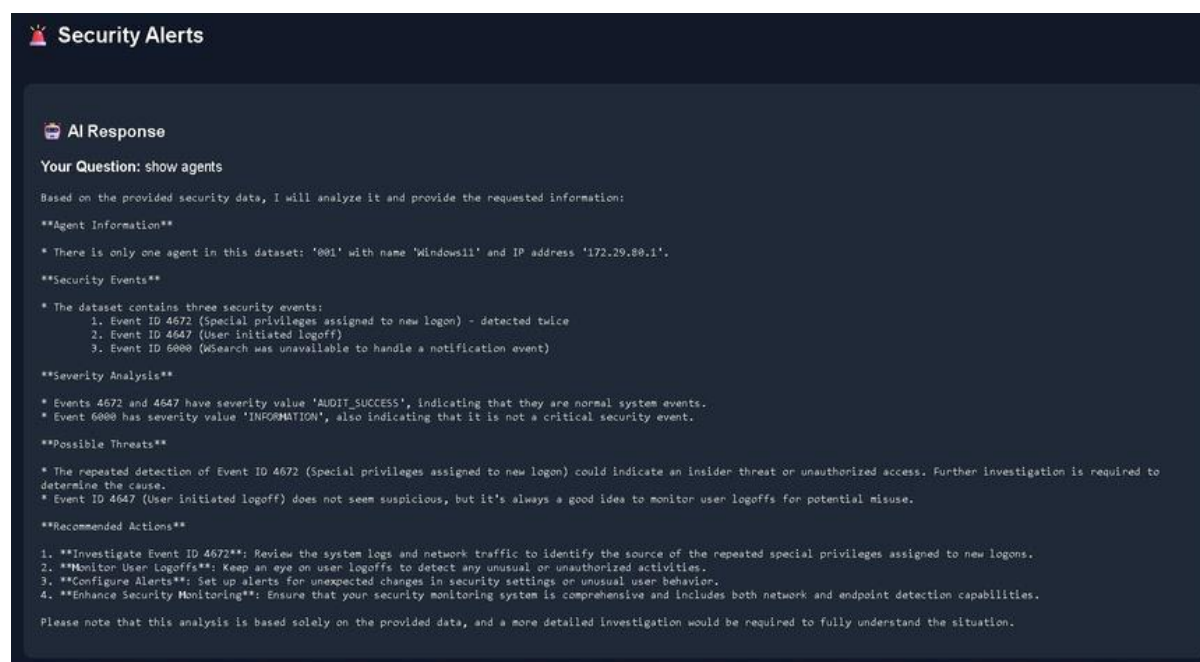


Figure 6.3.1: AI-powered chatbot providing automated cybersecurity analysis.

- **Show Splunk Alerts**

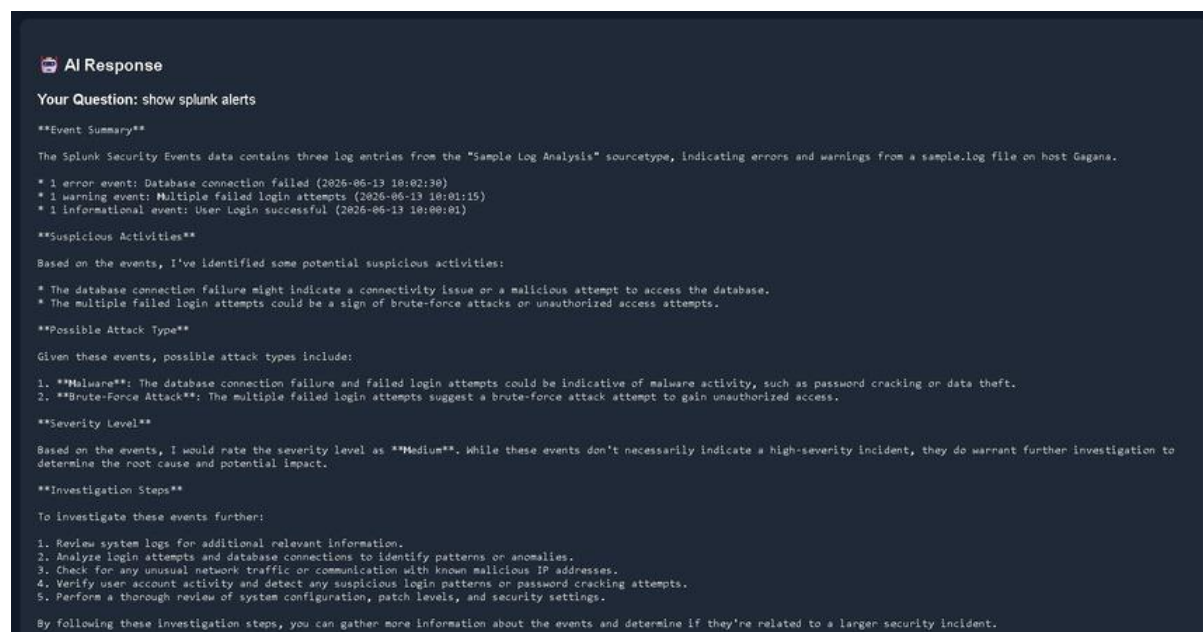


Figure 6.3.2: AI-powered chatbot providing automated cybersecurity analysis.

- **Investigate Alerts**

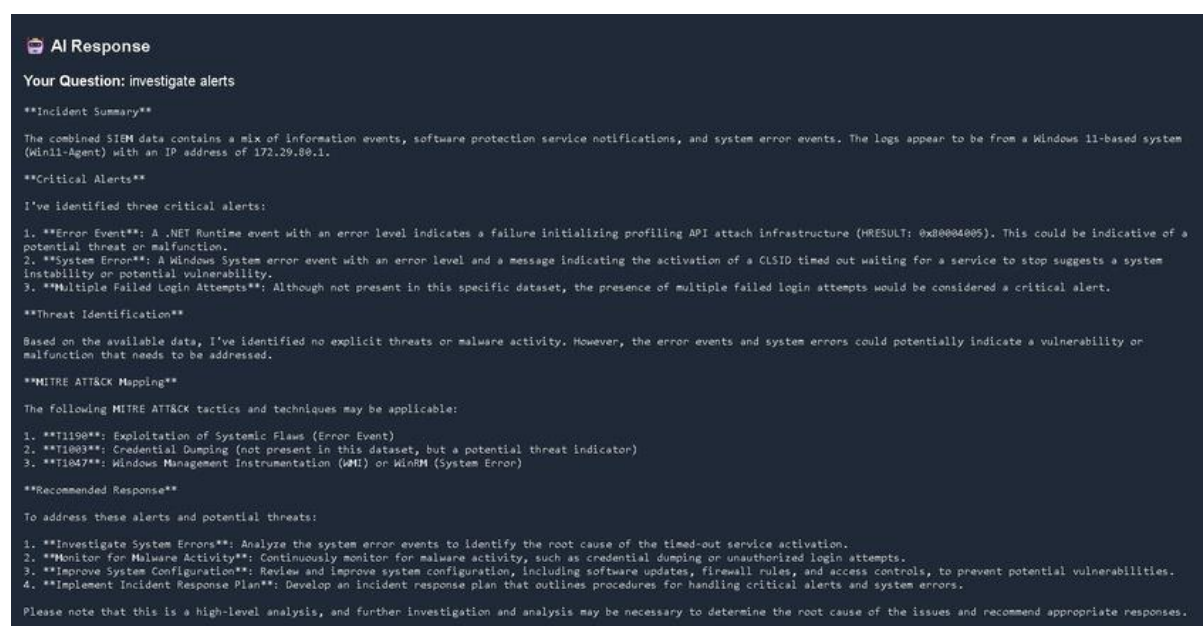


Figure 6.3.3: AI-powered chatbot providing automated cybersecurity analysis.

- **Explain Security Events**

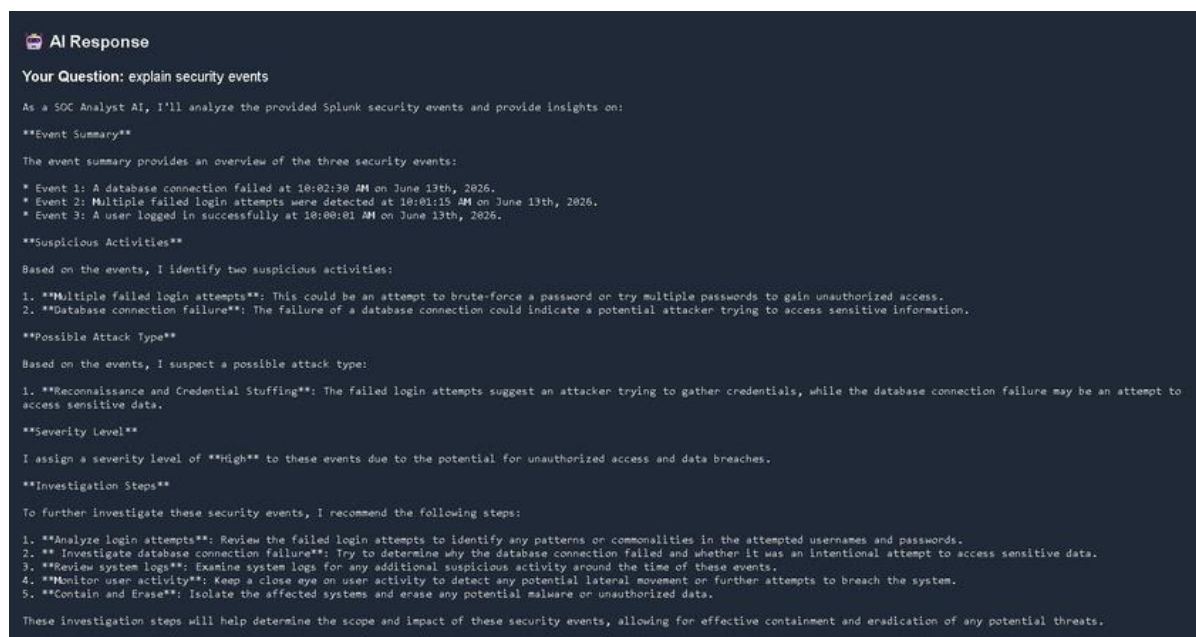


Figure 6.3.4: AI-powered chatbot providing automated cybersecurity analysis.

6.4 AI Alert Investigation

Each security alert can be individually investigated using the built-in AI Investigation module. When an analyst selects an alert, the chatbot generates a comprehensive incident report including:

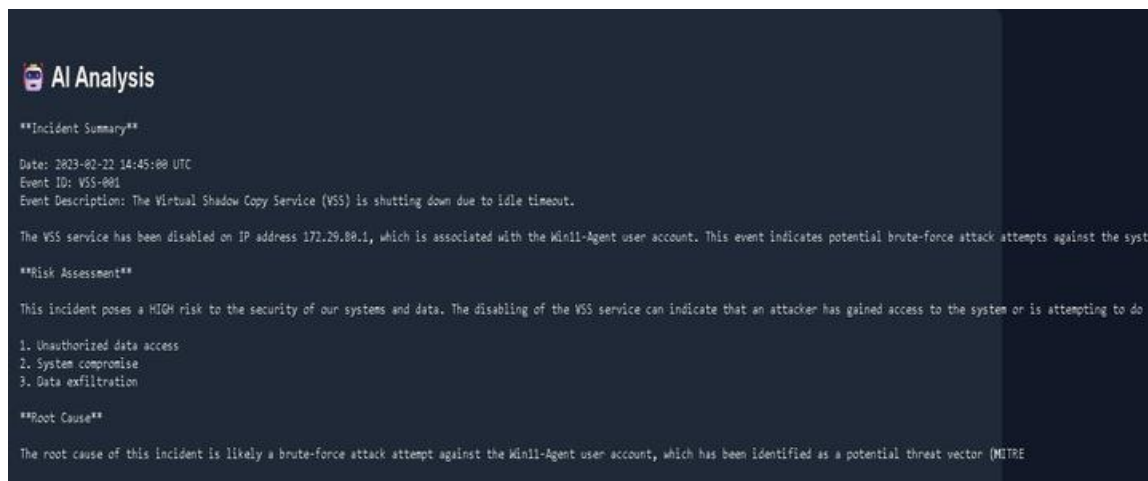
- Incident Summary •

Risk Assessment

- Root Cause Analysis
- Recommended Mitigation
- Security Recommendations

This functionality reduces the workload of SOC analysts by automatically converting raw log entries into understandable incident reports.

- **Incident Summary**



6.5 Splunk Log Analysis

Splunk Enterprise was successfully integrated to perform centralized log collection and event searching. Sample log events were indexed into Splunk and retrieved using Search Processing Language (SPL) queries.

The system successfully detected informational messages, warning events, and error logs including failed login attempts and database connection failures. These events were later displayed inside the Flask dashboard for unified analysis.

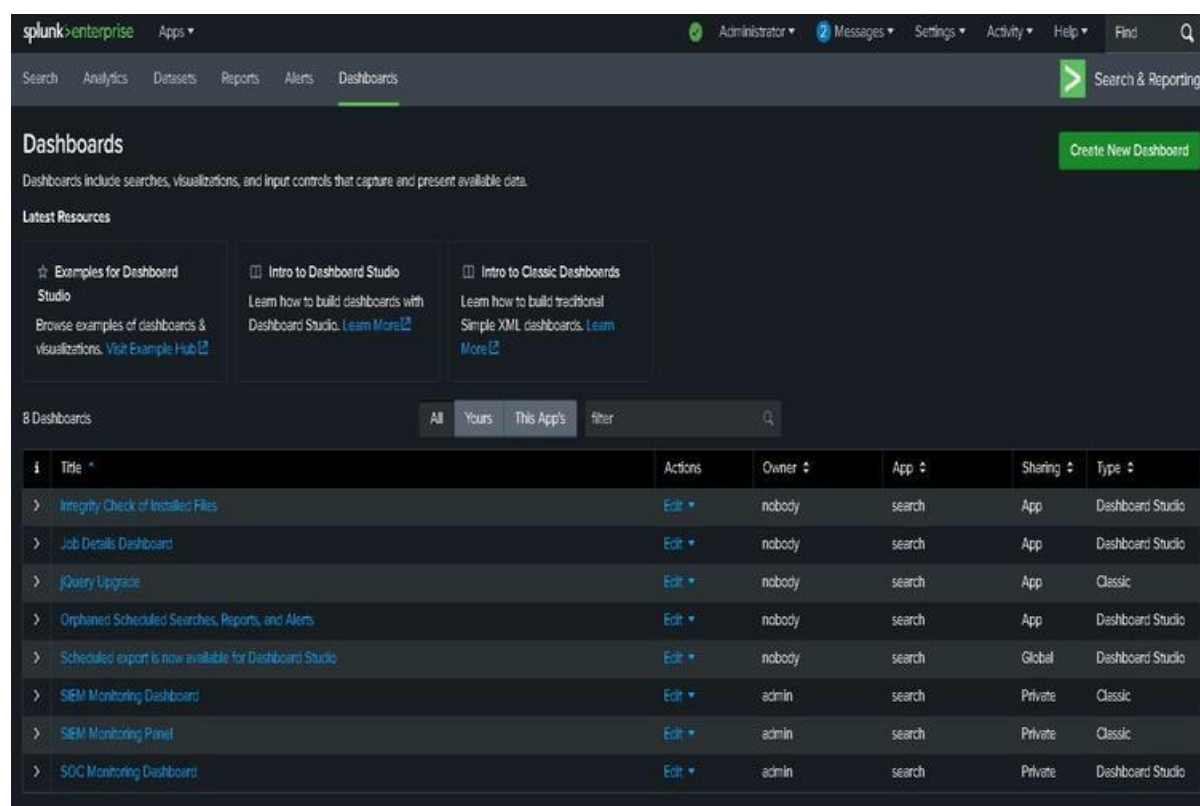


Figure 6.5: Splunk Enterprise displaying indexed security events.

6.6 PDF Report Generation

The system supports automatic PDF report generation for security auditing and documentation purposes. The generated report includes security statistics, alert summaries, top attacker information, security score, and complete alert details.

The PDF report provides an organized format suitable for management review and incident documentation.

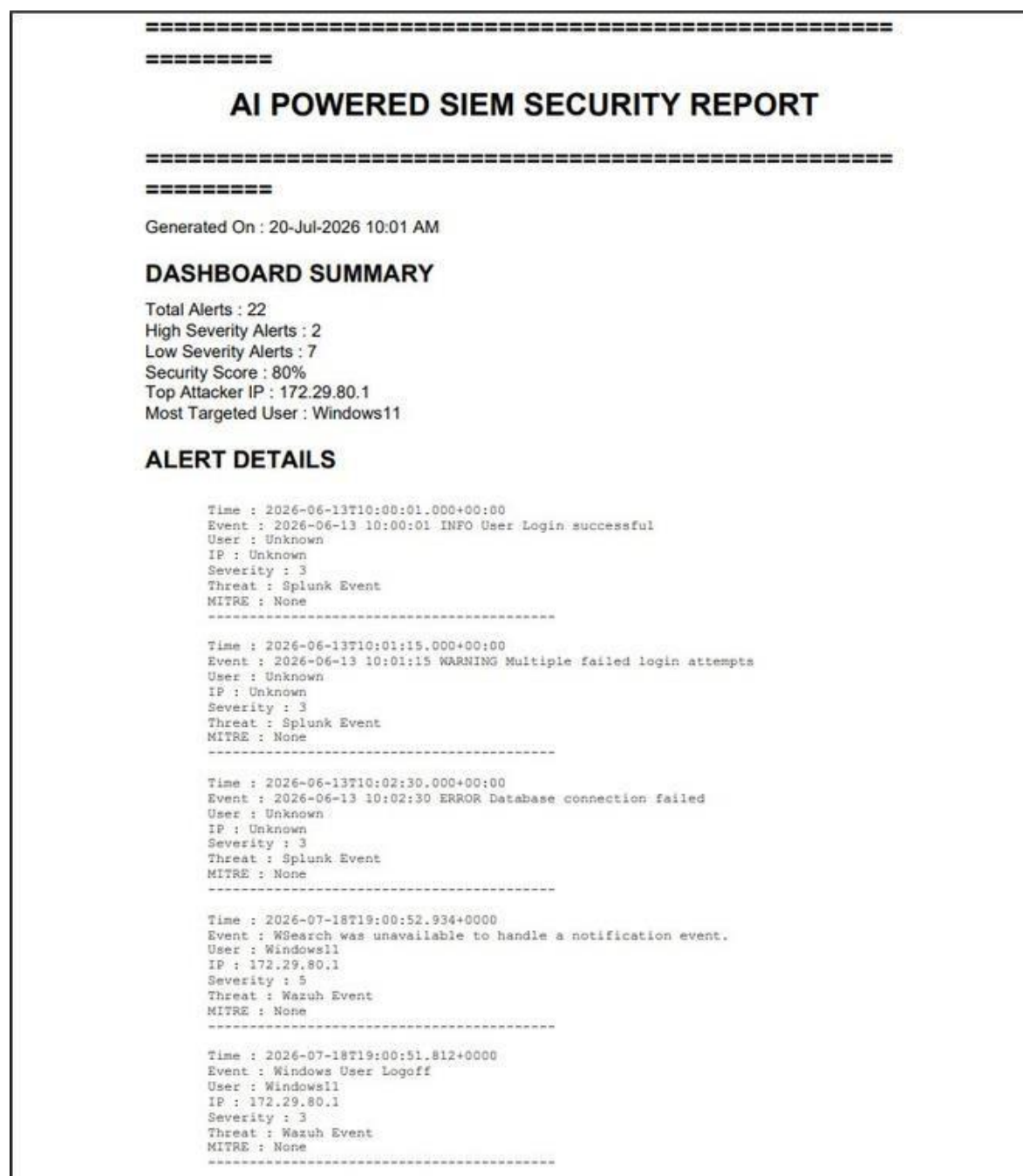


Figure 6.6: Automatically generated PDF security report.

6.7 CSV Export

The developed system also provides CSV export functionality, allowing security analysts to download all security alerts for further processing, archival, or external analysis using spreadsheet applications.

The exported file contains alert timestamps, event descriptions, users, IP addresses, severity levels, threat classifications, and MITRE ATT&CK mappings.

	A	B	C	D	E	F	G
1	Time	Event	User	IP	Severity	Threat	MITRE
2	#####	The VSS se	Win11-Age	172.29.80.	HIGH	BRUTE_FC	T1110
3	#####	Windows S	Win11-Age	172.29.80.	LOW	NORMAL	N/A
4	#####	Windows S	Win11-Age	172.29.80.	LOW	NORMAL	N/A
5	#####	Test Wazu	testuser	10.10.10.5	LOW	NORMAL	N/A
6	#####	Multiple fa	admin	192.168.1.	LOW	NORMAL	N/A
7	#####	Multiple fa	admin	192.168.1.	LOW	NORMAL	N/A
8	#####	Multiple fa	admin	192.168.1.	HIGH	BRUTE_FC	T1110
9	#####	Multiple fa	admin	192.168.1.	LOW	NORMAL	N/A
10	#####	Multiple fa	admin	192.168.1.	LOW	NORMAL	N/A
11							

Figure 6.7: CSV export containing complete security alert information.

6.8 Overall System Performance

The implemented solution successfully achieved the objectives defined at the beginning of the project. Security events generated from Wazuh and Splunk were successfully consolidated into a unified dashboard. Artificial intelligence significantly improved incident investigation by automatically interpreting security alerts and providing contextual explanations.

The integration of Flask, Wazuh, Splunk, SQLite, and the Llama 3 language model resulted in an intelligent Security Information and Event Management solution

capable of assisting Security Operations Center analysts in detecting, analyzing, and responding to cybersecurity incidents efficiently.

The generated dashboard, AI chatbot, automated investigation reports, and export capabilities demonstrate that the proposed solution provides an effective, scalable, and user-friendly approach for modern security monitoring environments.

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

The AI-Powered SIEM Alert Investigation Chatbot was successfully designed and implemented to enhance security monitoring and incident investigation processes. The proposed system integrates Wazuh and Splunk to collect, analyze, and centralize security events from monitored endpoints and log sources.

The developed Flask-based Security Operations Center dashboard provides a unified platform for monitoring alerts, visualizing security metrics, and

investigating potential threats. The integration with Wazuh enables endpoint monitoring, rule-based threat detection, and MITRE ATT&CK-based security

analysis, while Splunk provides centralized log management and event searching capabilities.

The implementation of a locally deployed Llama 3 Large Language Model through Ollama enables AI-assisted security analysis using natural language queries. The

chatbot reduces the manual effort required by SOC analysts by automatically summarizing alerts, identifying possible threats, performing incident analysis, and

suggesting recommended security actions.

The system also provides automated report generation through PDF and CSV export features, improving incident documentation and security reporting. The

successful integration of SIEM technologies with artificial intelligence demonstrates an effective approach toward improving cybersecurity operations

and reducing incident response time.

The proposed system integrates **Wazuh** and **Splunk Enterprise** to collect, process, and analyze security events from monitored endpoints and centralized log

sources. Wazuh provides endpoint monitoring, security event detection, rule-based analysis, and MITRE ATT&CK threat mapping, while Splunk enables efficient

log indexing, searching, and centralized event analysis.

A Flask-based web application was developed to provide a unified SOC dashboard that displays security alerts, threat information, visualization charts, and

important security metrics. The dashboard helps security analysts quickly understand the current security posture and identify potential security incidents.

The integration of Artificial Intelligence using the locally deployed **Llama 3 Large Language Model through Ollama** significantly enhances the investigation process. The AI chatbot allows analysts to interact with security data using natural language queries and provides automated explanations, incident summaries, risk assessments, and recommended response actions. This reduces the dependency on manual log analysis and improves the efficiency of security investigations.

The system also includes automated security report generation through PDF and CSV export functionality, enabling effective incident documentation, auditing, and

further analysis. The successful implementation demonstrates that combining SIEM technologies with Artificial Intelligence can provide a more intelligent,

efficient, and user-friendly approach to cybersecurity monitoring.

Overall, the proposed solution provides a scalable, intelligent, and user-friendly platform that assists security teams in detecting, analyzing, and responding to cybersecurity incidents efficiently.

The developed solution achieves the primary objectives of:

- Centralized security alert monitoring
- Integration of multiple SIEM platforms
- AI-assisted incident investigation
- Automated security analysis
- Improved SOC analyst efficiency

Therefore, the proposed AI-Powered SIEM Alert Investigation Chatbot provides an effective framework for enhancing modern cybersecurity operations.

7.2 Future Scope

Although the implemented system successfully achieves its objectives, several improvements can be introduced in future development:

1. Real-Time Threat Intelligence Integration

The system can be enhanced by integrating additional threat intelligence platforms such as VirusTotal, AbuseIPDB, and other security feeds to improve IP reputation analysis and threat identification.

2. Automated Incident Response

Future versions can include Security Orchestration, Automation, and Response (SOAR) capabilities to automatically perform actions such as:

- Blocking malicious IP addresses
- Disabling compromised accounts
- Isolating infected endpoints
- Creating automated incident tickets

3. Cloud Deployment

The application can be deployed on cloud platforms to support distributed security monitoring across multiple organizations and remote endpoints.

4. Advanced Machine Learning Detection

Machine learning models can be added to identify abnormal user behavior, detect unknown attacks, and predict potential security incidents.

5. Multi-Model AI Support

Future implementations can support multiple Large Language Models to improve accuracy, response quality, and domain-specific cybersecurity analysis.

6. Mobile Application Integration

A mobile-based SOC monitoring application can be developed to provide security alerts and AI investigation capabilities remotely.

7. Machine Learning-Based Threat Detection

Machine learning algorithms can be integrated to detect abnormal patterns and unknown threats. Behavioral analysis models can identify suspicious activities that may not match existing security rules.

8. Multi-Agent AI Security System

Future improvements can include multiple specialized AI agents for different cybersecurity tasks such as:

- Log analysis agent
- Threat intelligence agent
- Incident response agent
- Compliance monitoring agent

This would improve the accuracy and efficiency of security investigations.

7. Enhanced User Authentication and Access Control

Future implementations can include advanced security mechanisms such as:

- Role-Based Access Control (RBAC)
- Multi-Factor Authentication (MFA)
- User activity auditing

to improve the security of the SOC platform itself.

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APPENDIX

APPENDIX A

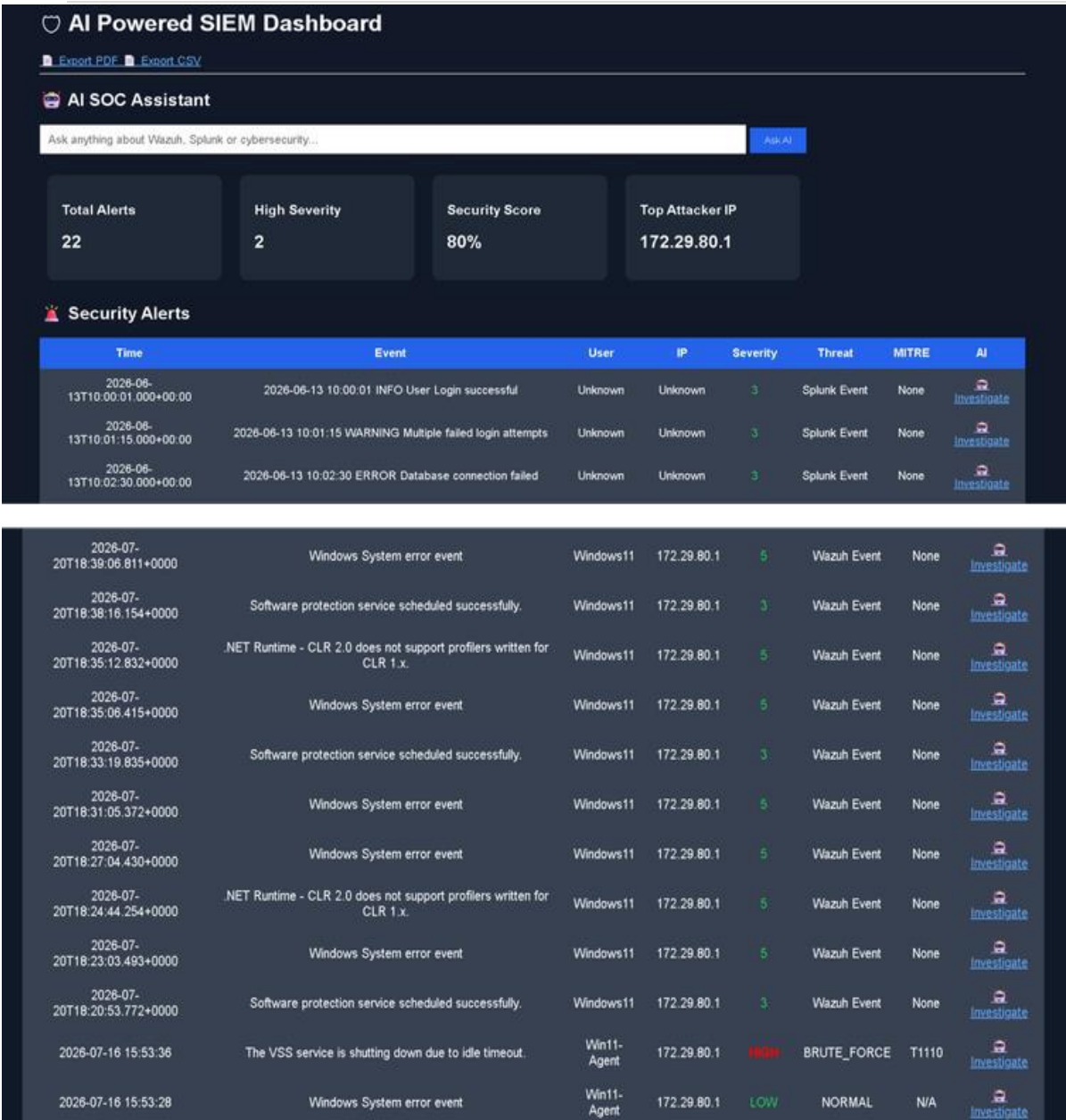
System Screenshots

This appendix presents the screenshots captured during the implementation and testing of the **AI-Powered SIEM Alert Investigation Chatbot using Wazuh and Splunk**. These screenshots demonstrate the successful integration of the SIEM platforms, AI investigation module, and report generation features.

Figure A.1 – AI-Powered SIEM Dashboard

The Flask-based dashboard displaying centralized security alerts collected from Wazuh and Splunk. It provides information such as alert severity, threat category, MITRE ATT&CK mapping, and AI investigation options.

Figure A.2 – Wazuh Dashboard



The Wazuh Dashboard showing the monitored endpoint and overall security monitoring status.

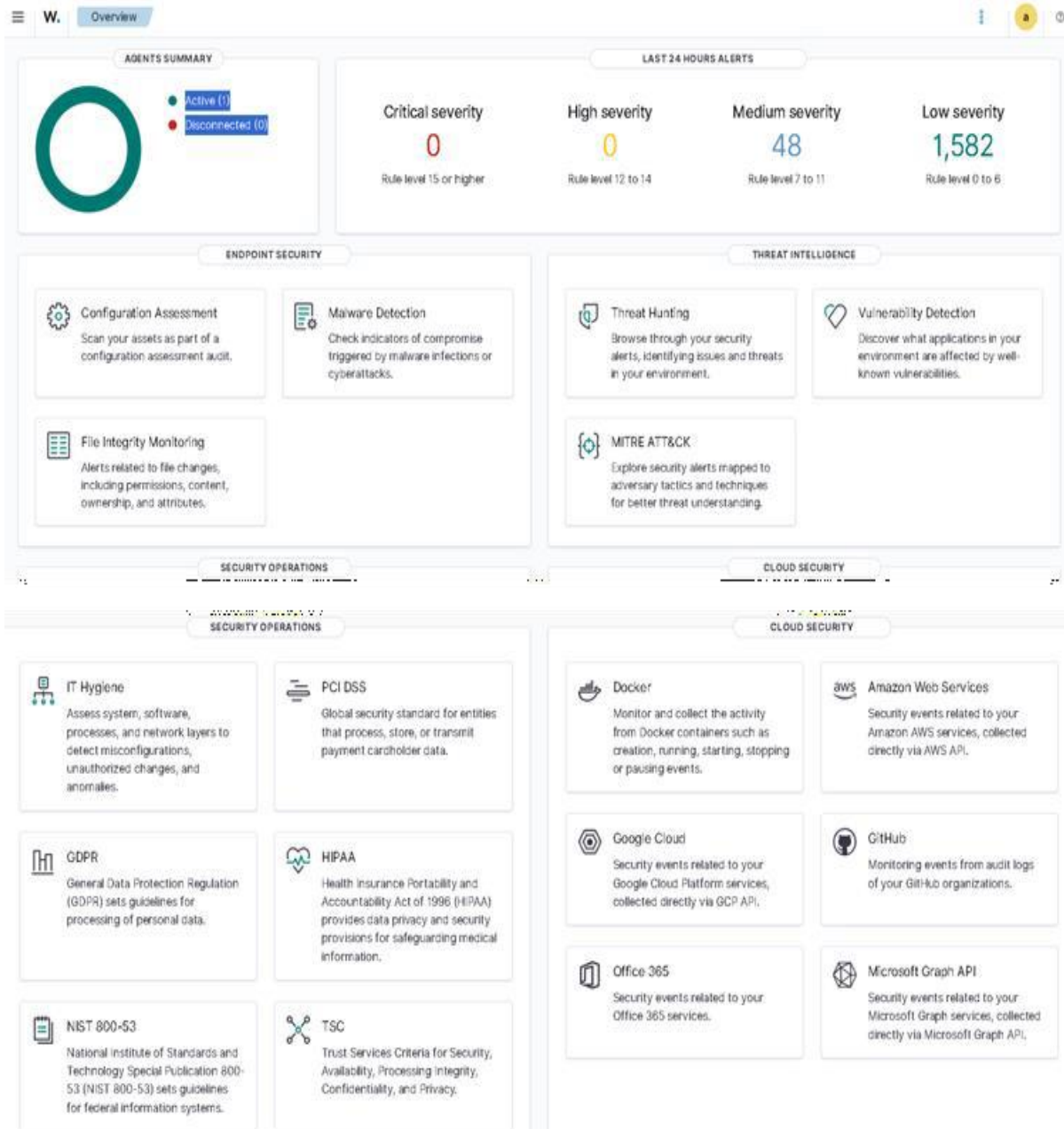
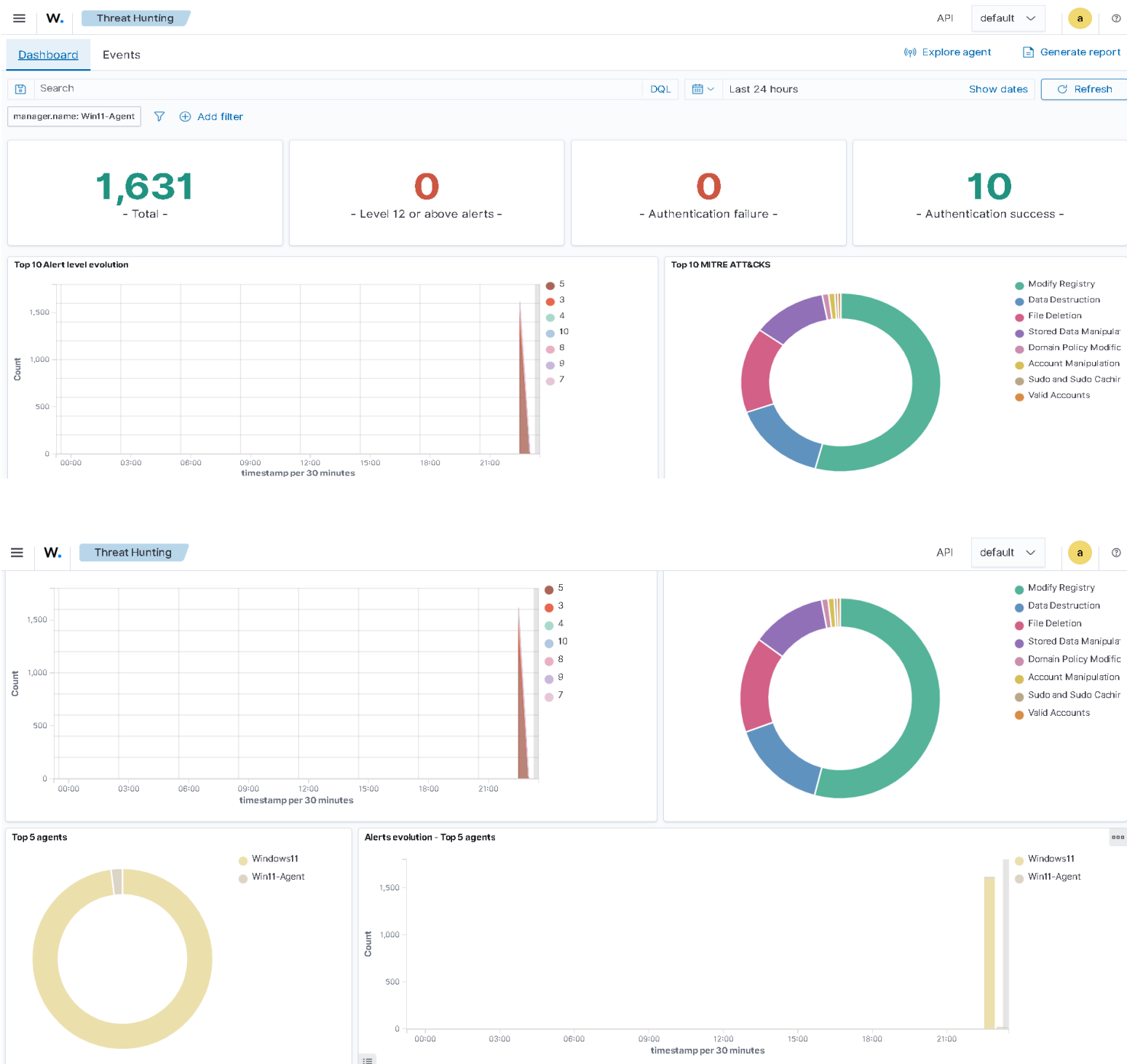


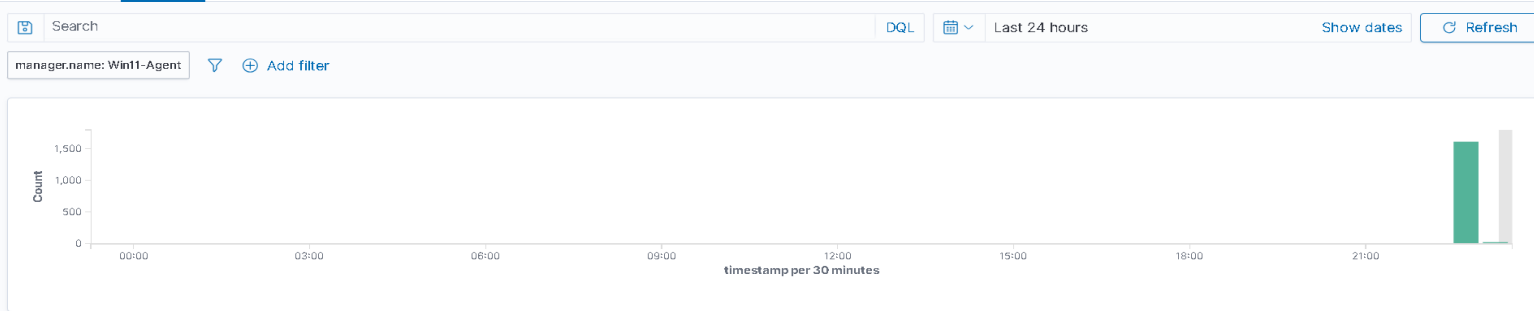
Figure A.3 – Wazuh Threat Hunting Dashboard

The Threat Hunting dashboard displaying security events, attack trends, and threat visualization generated by Wazuh.



Dashboard

Events



1,631 hits

Jul 19, 2026 @ 23:16:15.300 - Jul 20, 2026 @ 23:16:15.300

Export FormattedReset view789 available fieldsColumnsDensity1 fields sortedFull screen

timestamp	agent.name	rule.description	rule.le...	rule.id
Jul 20, 2026 @ 23:12:52.7...	Windows11	Windows System error event	5	61102
Jul 20, 2026 @ 23:09:11.4...	Win11-Agent	PAM: Login session closed.	3	5502
Jul 20, 2026 @ 23:08:53.9...	Win11-Agent	Wazuh server started.	3	502
Jul 20, 2026 @ 23:08:51.2...	Windows11	Windows System error event	5	61102
Jul 20, 2026 @ 23:08:24.9...	Win11-Agent	PAM: Login session opened.	3	5501

1,631 hits

Jul 19, 2026 @ 23:16:15.300 - Jul 20, 2026 @ 23:16:15.300

Export FormattedReset view789 available fieldsColumnsDensity1 fields sortedFull screen

timestamp	agent.name	rule.description	rule.le...	rule.id
Jul 20, 2026 @ 23:12:52.7...	Windows11	Windows System error event	5	61102
Jul 20, 2026 @ 23:09:11.4...	Win11-Agent	PAM: Login session closed.	3	5502
Jul 20, 2026 @ 23:08:53.9...	Win11-Agent	Wazuh server started.	3	502
Jul 20, 2026 @ 23:08:51.2...	Windows11	Windows System error event	5	61102
Jul 20, 2026 @ 23:08:24.9...	Win11-Agent	PAM: Login session opened.	3	5501
Jul 20, 2026 @ 23:08:24.9...	Win11-Agent	Successful sudo to ROOT executed.	3	5402
Jul 20, 2026 @ 23:06:46.9...	Win11-Agent	PAM: Login session closed.	3	5502
Jul 20, 2026 @ 23:06:38.9...	Win11-Agent	Successful sudo to ROOT executed.	3	5402
Jul 20, 2026 @ 23:06:38.9...	Win11-Agent	PAM: Login session opened.	3	5501
Jul 20, 2026 @ 23:06:36.9...	Win11-Agent	PAM: Login session closed.	3	5502
Jul 20, 2026 @ 23:04:50.8...	Windows11	Windows System error event	5	61102
Jul 20, 2026 @ 23:03:52.9...	Win11-Agent	PAM: Login session opened.	3	5501
Jul 20, 2026 @ 23:03:52.9...	Win11-Agent	Successful sudo to ROOT executed.	3	5402
Jul 20, 2026 @ 23:02:42.8...	Win11-Agent	Successful sudo to ROOT executed.	3	5402
Jul 20, 2026 @ 23:02:42.8...	Win11-Agent	PAM: Login session opened.	3	5501

Figure A.4– Splunk Search Dashboard

The Splunk Enterprise search interface demonstrating indexed security logs and search queries used for log analysis.

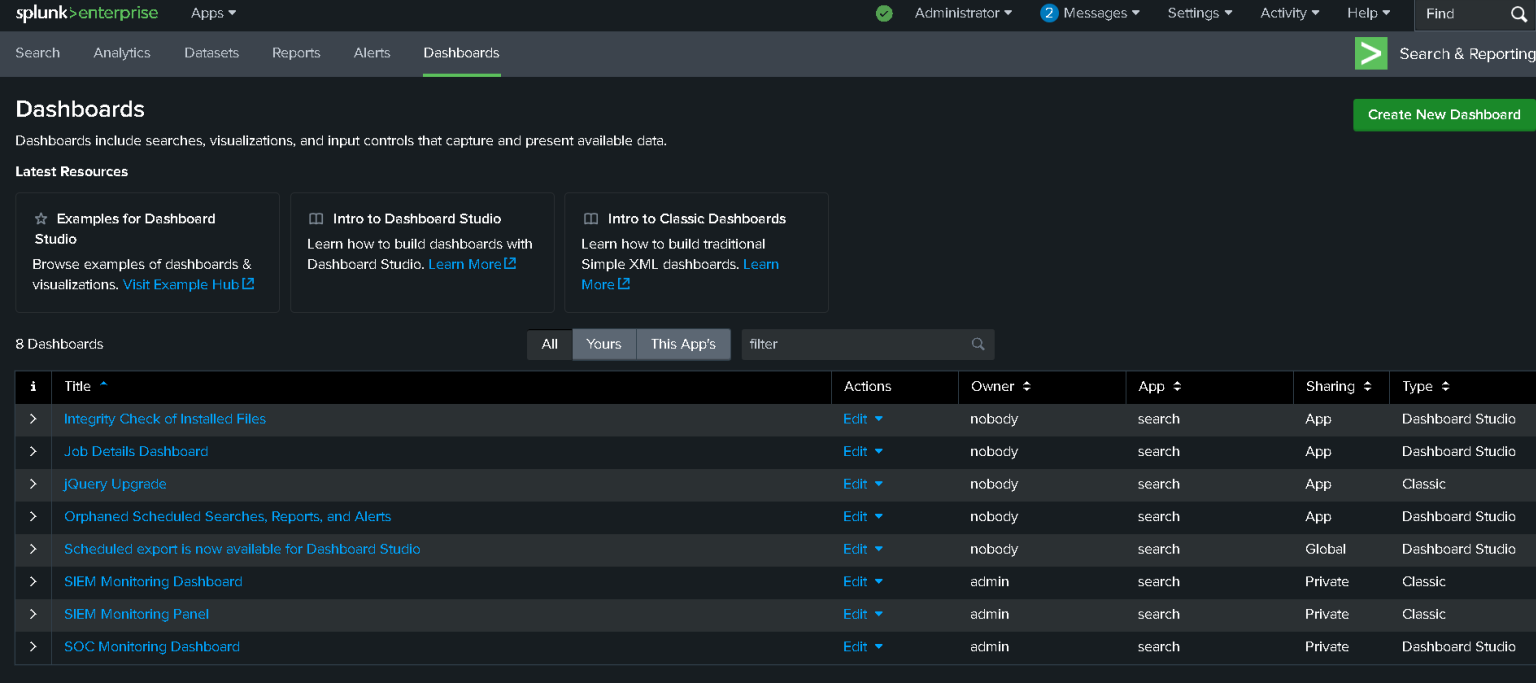


Figure A.5— AI Chatbot Response

The AI chatbot interface demonstrating natural language interaction for security alert investigation using the Llama 3 model.

 Security Alerts

 AI Response

Your Question: show agents

Based on the provided security data, I will analyze it and provide the requested information:

Agent Information

* There is only one agent in this dataset: '001' with name 'Windows11' and IP address '172.29.80.1'.

Security Events

* The dataset contains three security events:

- 1. Event ID 4672 (Special privileges assigned to new logon) - detected twice
- 2. Event ID 4647 (User initiated logoff)
- 3. Event ID 6000 (WSearch was unavailable to handle a notification event)

Severity Analysis

* Events 4672 and 4647 have severity value 'AUDIT_SUCCESS', indicating that they are normal system events.

* Event 6000 has severity value 'INFORMATION', also indicating that it is not a critical security event.

Possible Threats

* The repeated detection of Event ID 4672 (Special privileges assigned to new logon) could indicate an insider threat or unauthorized access. Further investigation is required to determine the cause.

* Event ID 4647 (User initiated logoff) does not seem suspicious, but it's always a good idea to monitor user logoffs for potential misuse.

Recommended Actions

- 1. **Investigate Event ID 4672**: Review the system logs and network traffic to identify the source of the repeated special privileges assigned to new logons.
- 2. **Monitor User Logoffs**: Keep an eye on user logoffs to detect any unusual or unauthorized activities.
- 3. **Configure Alerts**: Set up alerts for unexpected changes in security settings or unusual user behavior.
- 4. **Enhance Security Monitoring**: Ensure that your security monitoring system is comprehensive and includes both network and endpoint detection capabilities.

Please note that this analysis is based solely on the provided data, and a more detailed investigation would be required to fully understand the situation.

 AI Response

Your Question: show splunk alerts

Event Summary

The Splunk Security Events data contains three log entries from the "Sample Log Analysis" sourcetype, indicating errors and warnings from a sample.log file on host Gagana.

* 1 error event: Database connection failed (2026-06-13 10:02:30)

* 1 warning event: Multiple failed login attempts (2026-06-13 10:01:15)

* 1 informational event: User Login successful (2026-06-13 10:00:01)

Suspicious Activities

Based on the events, I've identified some potential suspicious activities:

* The database connection failure might indicate a connectivity issue or a malicious attempt to access the database.

* The multiple failed login attempts could be a sign of brute-force attacks or unauthorized access attempts.

Possible Attack Type

Given these events, possible attack types include:

- 1. **Malware**: The database connection failure and failed login attempts could be indicative of malware activity, such as password cracking or data theft.
- 2. **Brute-Force Attack**: The multiple failed login attempts suggest a brute-force attack attempt to gain unauthorized access.

Severity Level

Based on the events, I would rate the severity level as **Medium**. While these events don't necessarily indicate a high-severity incident, they do warrant further investigation to determine the root cause and potential impact.

Investigation Steps

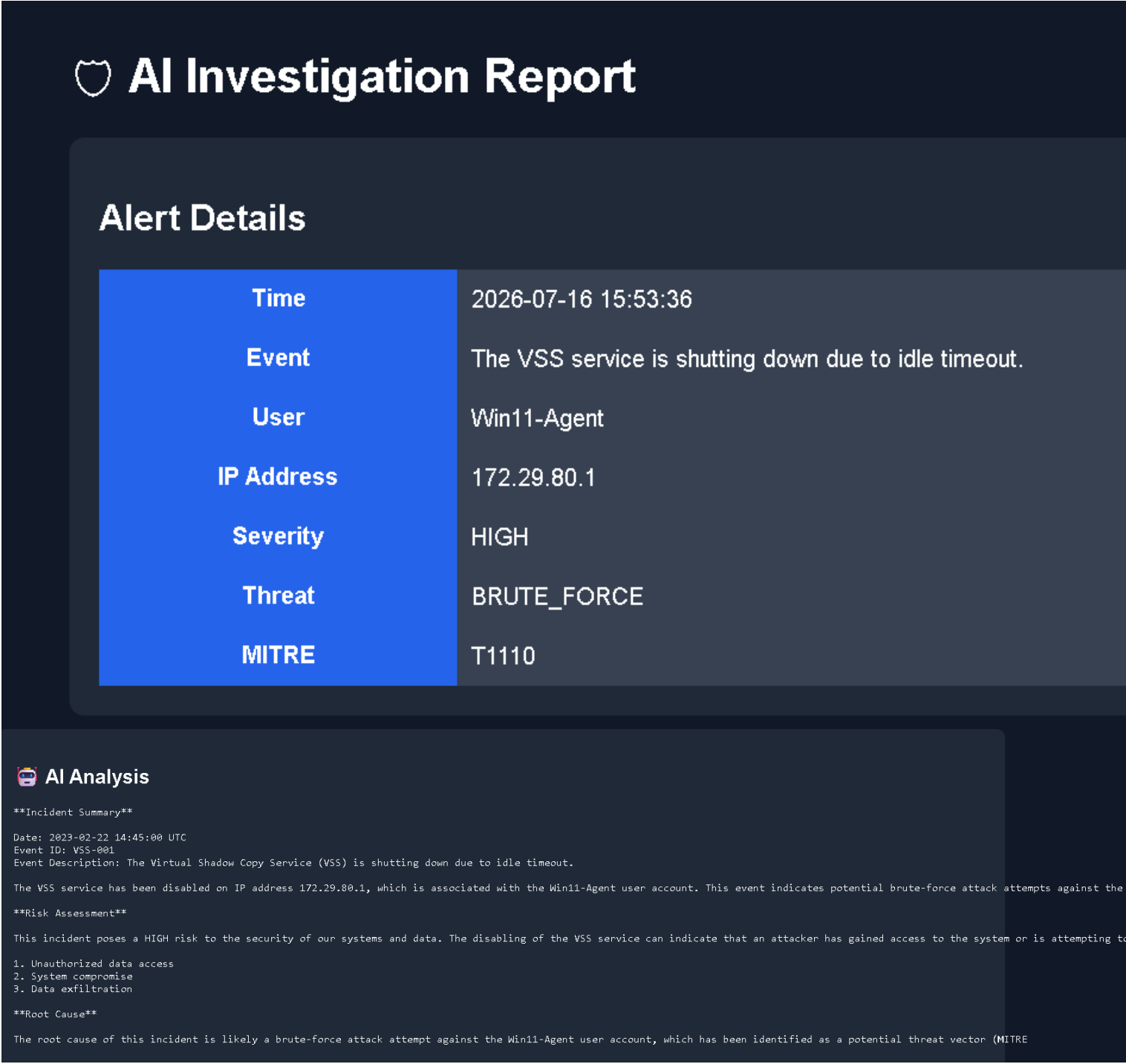
To investigate these events further:

- 1. Review system logs for additional relevant information.
- 2. Analyze login attempts and database connections to identify patterns or anomalies.
- 3. Check for any unusual network traffic or communication with known malicious IP addresses.
- 4. Verify user account activity and detect any suspicious login patterns or password cracking attempts.
- 5. Perform a thorough review of system configuration, patch levels, and security settings.

By following these investigation steps, you can gather more information about the events and determine if they're related to a larger security incident.

Figure A.6 – AI Investigation Report

The AI-generated investigation report providing an incident summary, risk assessment, root cause analysis, and recommended response.





Threat Intelligence

No Threat Intelligence Available.

[← Back to Dashboard](#)

Figure A.7– PDF Security Report

The generated PDF report containing summarized security alerts, statistics, and incident information.

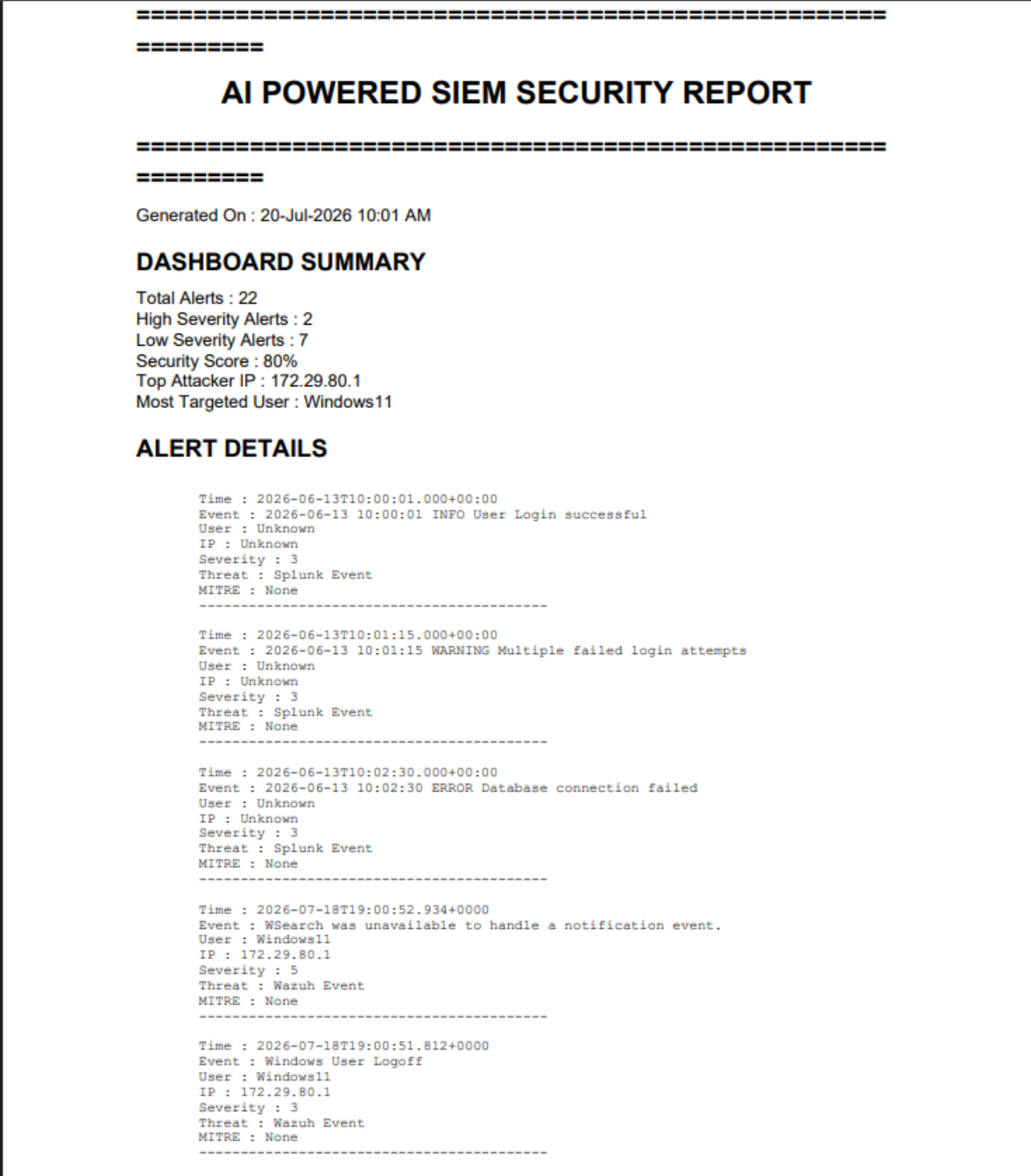


Figure A.8 – CSV Alert Export

The generated CSV file containing exported security alert information for further analysis.

	A	B	C	D	E	F	G
1	Time	Event	User	IP	Severity	Threat	MITRE
2	#####	The VSS se	Win11-Age	172.29.80.	HIGH	BRUTE_FC	T1110
3	#####	Windows S	Win11-Age	172.29.80.	LOW	NORMAL	N/A
4	#####	Windows S	Win11-Age	172.29.80.	LOW	NORMAL	N/A
5	#####	Test Wazu	testuser	10.10.10.5	LOW	NORMAL	N/A
6	#####	Multiple fa	admin	192.168.1.	LOW	NORMAL	N/A
7	#####	Multiple fa	admin	192.168.1.	LOW	NORMAL	N/A
8	#####	Multiple fa	admin	192.168.1.	HIGH	BRUTE_FC	T1110
9	#####	Multiple fa	admin	192.168.1.	LOW	NORMAL	N/A
10	#####	Multiple fa	admin	192.168.1.	LOW	NORMAL	N/A
11							

APPENDIX B

Configuration Details

B.1 Wazuh Configuration

The Wazuh platform was configured to monitor security events from the Windows endpoint and provide centralized threat detection.

Configuration Summary:

- Wazuh Manager Version: **4.14.6**
- Wazuh Agent Version: **4.14.6**
- Wazuh Dashboard: **OpenSearch Dashboards**
- Communication Protocol: **HTTPS REST API**
- Operating System: **Ubuntu 24.04 LTS**

B.2 Splunk Configuration

Splunk Enterprise was integrated to collect, index, and analyze security logs generated during the project.

Configuration Summary:

- Splunk Enterprise
- REST API Integration
- Log Index: main
- Output Format: JSON
- Search Processing: Search Jobs API

B.3 Flask Application Structure

The project was developed using the Flask web framework with a modular architecture.

AI-Powered-SIEM-Chatbot/

```
├─ app.py
├─ chatbot.py
├─ wazuh_api.py
├─ splunk_api.py
├─ database.py
├─ report.py
├─ templates/
├─ static/
├─ reports/
└─ alerts.db
```

B.4 Database Schema

The application stores processed alerts in an SQLite database.

Field	Data Type	Description
id	INTEGER	Alert Identifier
time	TEXT	Alert Timestamp
event	TEXT	Security Event
ip	TEXT	Source IP Address
user	TEXT	Username
severity	TEXT	Alert Severity
threat	TEXT	Threat Category
mitre	TEXT	MITRE ATT&CK Technique

B.5 AI Model Configuration

The AI investigation module was implemented using a locally hosted Large Language Model.

Configuration Summary:

- AI Runtime: Ollama
- AI Model: Llama 3
- Framework: Flask
- Response Format: JSON
- Purpose: AI-assisted security alert investigation and incident analysis